The University of Tokyo

Beam Propagation Simulation in SiO_2

Author
Anatole Cremel–Schlemer

Date
August 4, 2026

Contents

1 Introduction

Since their invention in the early 80's [1], femtosecond lasers have rapidly advanced the frontiers of materials science research due to their high peak intensity and ultra short temporal resolution.

The ultra-short nature of femtosecond pulses has enabled imaging of ultrafast processes, such as chemical reactions [2] and the complex energy transfer dynamics between electrons and crystal lattices in solids [3], whereas the high peak intensity of these pulses allows them to ionise matter and trigger laser-produced plasma (LPP), which emits a variety of spectra, including X-rays [4]. Among these nonlinear effects, Kerr self-focusing, plasma defocusing, and multiphoton/avalanche ionization play a critical role in the dynamics of femtosecond pulse propagation in dielectrics.

Studying the dynamics of femtosecond pulses in dielectric media, ranging from the near-UV to the near-infrared, helps clarify the transfer of electromagnetic energy to electrons, followed by electron-lattice interactions that alter the lattice and convert energy into heat [4]. However, such studies remain challenging, and some phenomena remain difficult to model for wide-bandgap dielectric materials such as silicon dioxide (SiO_2). In this study, we focus on SiO_2, a material widely used in optics due to its transparency and nonlinear properties.

Indeed, these media are transparent under low-intensity irradiation, and when they are irradiated by an intense femtosecond pulse, a cascade of effects begins, ranging from nonlinear ionisation occurring within a few femtoseconds to thermal effects occurring on picosecond to nanosecond timescales [5]. One of the most striking phenomena in this context is filamentation, where the beam undergoes self-focusing due to the Kerr effect, balanced by plasma defocusing, leading to the formation of stable light filaments. Once nonlinear ionisation becomes significant, the material loses its transparency, leading to photo-induced damage.

Femtosecond laser processing techniques have an edge over continuous-wave laser machining techniques because of the extremely fast interaction time between the beam and the sample. The damage induced by femtosecond laser pulses is mainly driven by electronic excitation and associated nonlinear processes before the material lattice has equilibrated with the excited carriers [4]. This allows for better control over collateral thermal damage and thermal stresses, which are typically associated with continuous laser processing. Also, the nonlinear response allows for structural modifications of dielectrics inside the bulk and paves the way for microstructuring that creates subwavelength "voxels" [4].

Theoretical models for femtosecond pulse propagation in dielectrics have been extensively developed [7]. The complexity of these nonlinear interactions has pushed for extensive research aimed at understanding their fundamental mechanisms and harnessing these properties to develop precision laser processing techniques.

This work presents a simulation of beam propagation in SiO_2 using the nonlinear Schrödinger equation and equations governing electron density evolution. We will explain the physics underlying the model, as well as some of the numerical methods used to solve this complex system.

2 Excitation of Free Carriers

Current experimental methods for generating femtosecond lasers in the near ultraviolet range (NUV) remain complex, which limits research on the linear optical transitions of wide bandgap dielectric materials such as SiO_2.

Given these experimental limitations, research has focused mainly on the study of nonlinear absorption processes that generate free carriers in the conduction band. By responding to the electric field, these electrons increase their energy and trigger effects such as avalanche ionisation, which modify the optical properties of the material, in particular its absorbance [4, 5].

In this section, we explain the nonlinear mechanisms responsible for the generation of free carriers, that is, the excitation of electrons from the valence band to the conduction band. In wide bandgap dielectrics, these carriers are central to the light matter interaction, since they are the only way to transfer optical energy to the crystal lattice of the material, which is otherwise transparent to low intensity light.

2.1 Photoionization

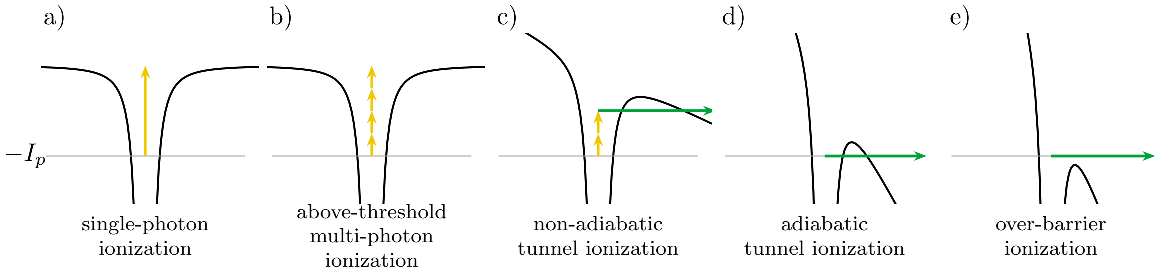


Figure 1: Illustration of the different ionization regimes.

Strong focusing leads to a high optical intensity in the material. This strong concentration of photons at the same location increases the probability of multiphoton ionization, a nonlinear effect [4]. The more intense the field, the more the potential binding the electron to the ion is deformed. Beyond a certain intensity threshold, the barrier takes a triangular shape and becomes thin enough to allow tunnel ionization. The Keldysh parameter was introduced to distinguish between the regime dominated by multiphoton ionization and the regime dominated by tunnel ionization [4].

In the semiclassical model, an electron escapes through a barrier of width $l = U_i/e\mathcal{E}$ with a velocity $v = \sqrt{2U_i/m}$. The time needed to cross this distance is:

$$\tau_{\text{tun}} = \frac{l}{v} = \frac{U_i/e\mathcal{E}}{\sqrt{2U_i/m}} = \frac{\sqrt{2mU_i}}{e\mathcal{E}} \tag{1}$$

The characteristic time during which tunnel ionization can occur is given by $\tau \approx 1/\omega_0$. The ratio of these two times is then:

$$\gamma = \frac{\tau_{\text{tun}}}{\tau} = \omega_0 \frac{\sqrt{2mU_i}}{e\mathcal{E}} \tag{2}$$

If the ratio $\gamma \ll 1$, the tunnel crossing time is very short compared to the ionization time. Tunnel ionization is then instantaneous on the scale of the laser cycle, and the field behaves as a quasi-static field. Conversely, if $\gamma \gg 1$, the field oscillates too fast for the electron to cross the barrier by tunneling, and ionization then requires multiphoton absorption [5].

The Keldysh photoionization rate is defined as follows:

$$W_{PI}(|\mathcal{E}|) = \frac{2\omega_0}{9\pi}\left(\frac{\omega_0 m}{\hbar\sqrt{\gamma_1}}\right)^{\frac{3}{2}} Q(\gamma, x)\exp(-\alpha\lceil x+1\rceil) \tag{3}$$

With $\gamma_1 = \frac{\gamma^2}{1+\gamma^2}$, $\gamma_2 = \frac{1}{1+\gamma^2}$ and $x = \frac{2}{\pi}\frac{U_i E(\gamma_2)}{\hbar\omega_0\sqrt{\gamma_1}}$

$$Q(\gamma, x) = \frac{\pi}{2K(\gamma_2)} \sum_{n=0}^{\infty} \exp\left(-n\pi\frac{K(\gamma_1)-E(\gamma_1)}{E(\gamma_2)}\right)\Phi\left(\frac{\pi}{2}\sqrt{\frac{n+2\lceil x+1\rceil-2x}{K(\gamma_2)E(\gamma_2)}}\right) \tag{4}$$

Φ is the Dawson function $\Phi(z) = e^{-z^2} \int_0^z e^{y^2} dy$ and K and E are respectively the complete elliptic integrals of the first and second kind, with the parameter $m = k^2$

$$K(m) = \int_0^{\pi/2} [1 - m\sin(t)^2]^{-1/2} dt \tag{5}$$

$$E(m) = \int_0^{\pi/2} [1 - m\sin(t)^2]^{1/2} dt \tag{6}$$

As observed in reference [6], filamentation in fused silica typically occurs at intensities around 5×10^{13} W/cm^2, where multiphoton and tunnel ionization both contribute significantly.

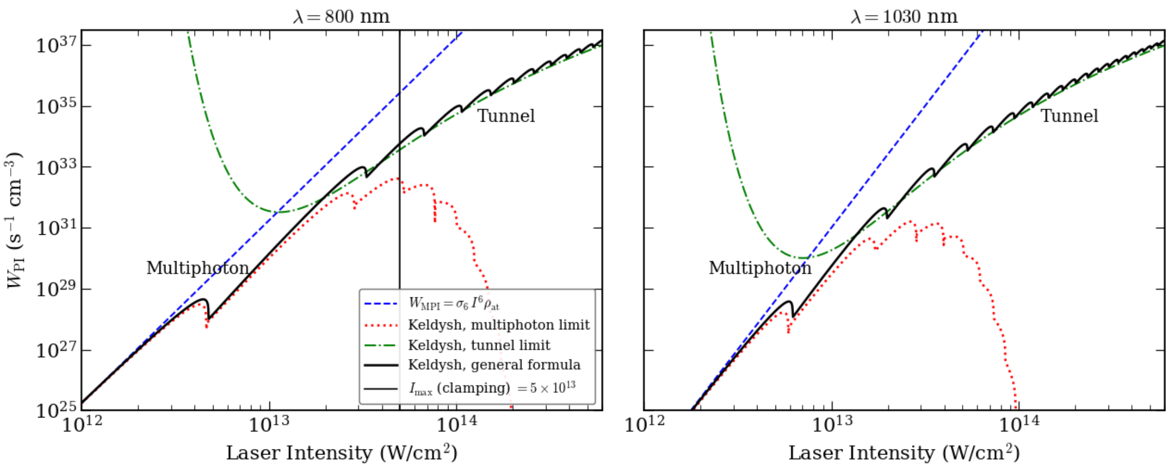


Figure 2: The photoionization rate $W_{\rm PI}$ in fused silica ($U_i = 9$ eV) is plotted against laser intensity for $\lambda = 800$ nm and $\lambda = 1030$ nm. The solid black line represents the full general Keldysh formula, while the dotted red and dash dotted green lines show its exact analytical asymptotes for the multiphoton limit ($\gamma \gg 1$) and the tunnel limit ($\gamma \ll 1$) respectively. The dotted blue line corresponds to the standard perturbative MPI law ($W_{\rm MPI} = \sigma_K I^K \rho_{\rm at}$). The vertical gray line indicates the typical clamping intensity level ($\sim 5 \times 10^{13}$ W/cm^2) reached in filamentation simulations at 800 nm.

2.2 Free Carrier Absorption

Once electrons are promoted to the conduction band, they behave as quasi free carriers. The laser field accelerates them, and they drift in opposition to the electrical field $\vec{E}$ [4]. One might expect these carriers to keep absorbing photons and gaining energy directly. However, single photon absorption by an isolated free electron is forbidden, because it cannot simultaneously satisfy conservation of energy and conservation of momentum.

For the absorption of a photon $(\omega_0, \vec{k}_{\rm ph})$ by an electron with initial wave vector $\vec{k}_i$ moving to a final state with wave vector $\vec{k}_f$, the laws of conservation of energy and momentum must be satisfied. They read:

$$E_f = E_i + \hbar\omega_0, \qquad \vec{k}_f = \vec{k}_i + \vec{k}_{\rm ph}. \tag{7}$$

Using the parabolic band $E = \hbar^2 k^2 / 2m^*$ and the photon momentum $k_{\rm ph} = \omega_0 / c = 2\pi/\lambda_0$, equation (7) becomes:

$$\frac{\hbar^2 k_f^2}{2m^*} = \frac{\hbar^2 k_i^2}{2m^*} + \hbar\omega_0, \qquad k_f = k_i + \frac{2\pi}{\lambda_0}. \tag{8}$$

The photon momentum is set by the wavelength alone. At $\lambda_0 = 1030$ nm, $k_{\text{ph}} = 2\pi/\lambda_0 \approx 6.1 \times 10^6\ \text{m}^{-1}$. The electron momentum, on the other hand, is set by its kinetic energy in the conduction band. Taking a representative value $E_i \sim 1$ eV, and any energy of order one eV leads to the same conclusion, we obtain:

$$k_i = \frac{\sqrt{2m^* E_i}}{\hbar} \approx 5 \times 10^9\ \text{m}^{-1}, \qquad \frac{k_{\text{ph}}}{k_i} \approx 10^{-3}. \tag{9}$$

Conservation of momentum therefore forces $k_f \approx k_i$, hence $E_f \approx E_i$, which contradicts $E_f = E_i + \hbar\omega_0$ unless $\hbar\omega_0 \approx 0$.

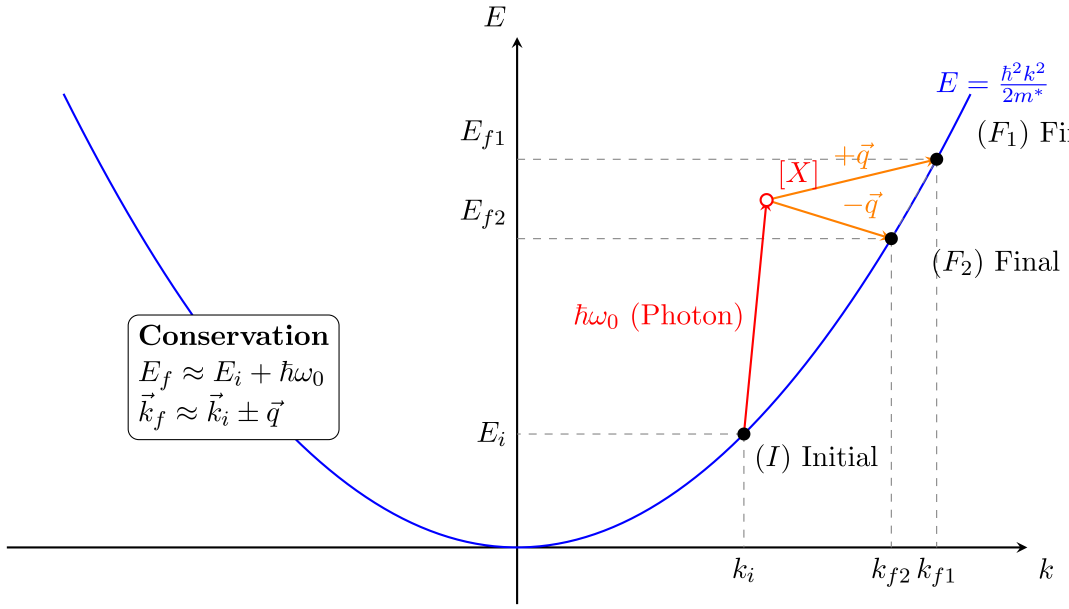
The process becomes allowed if a third body carries away the momentum mismatch. In a polar dielectric such as SiO_2, this third body is a phonon of wave vector $\vec{q}$ and energy $\hbar\omega_q$, so that:

$$E_f = E_i + \hbar\omega_0 \pm \hbar\omega_q, \qquad \vec{k}_f = \vec{k}_i + \vec{k}_{\text{ph}} \pm \vec{q}. \tag{10}$$

Since phonon wave vectors span the entire Brillouin zone:

$$q \lesssim \frac{\pi}{a} \sim 10^{10}\ \text{m}^{-1} \gtrsim k_i$$

Momentum can now be balanced, while $\hbar\omega_q \sim 0.06$–0.15 eV $\ll \hbar\omega_0$ leaves the energy balance essentially unchanged. This phonon assisted mechanism is free carrier absorption (FCA), equivalently inverse Bremsstrahlung [4, 5].



2.3 Avalanche Ionization

The energy shift is given by $\Delta E = \hbar\omega_0 \pm \hbar\omega_q$, where $\hbar\omega_q$ is the phonon energy and $\hbar\omega_0$ is the photon energy. Since the phonon energy is very small compared to the photon energy, we can assume $\Delta E \approx \hbar\omega_0$. Electrons can absorb several photons in sequence, increasing their velocity and their energy. Beyond a certain threshold, the energy of the free carriers exceeds the gap energy, and they can then collide with electrons of the valence band, causing impact ionization. This leaves two electrons in the conduction band, which increases the absorption efficiency and triggers further impact ionization events. This cascading effect is known as avalanche ionization. It combines free carrier absorption with impact ionization, sketched in Figure 3.

This process can repeat itself as long as the electric field is present and intense enough.

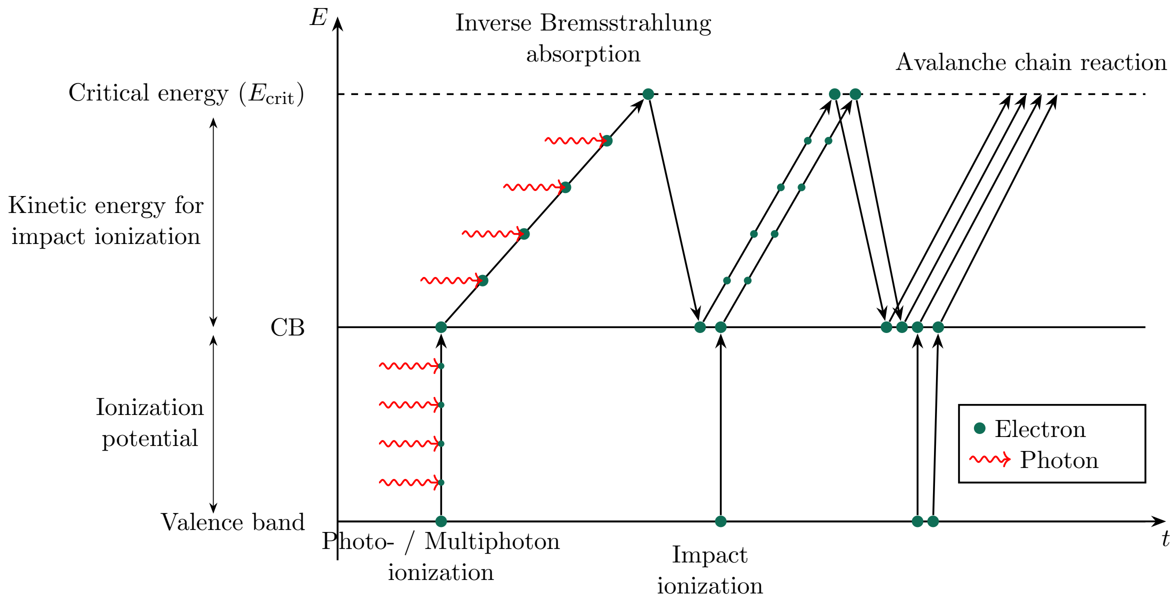


Figure 3: Avalanche chain reaction. A seed electron is promoted to the conduction band by photoionization, heated above the critical energy E_{crit} by successive inverse Bremsstrahlung absorption, and knocks a second valence electron into the conduction band by impact ionization. Both electrons repeat the cycle, doubling the population at each generation.

2.4 Optical Kerr Effect and Self-Steepening

When a laser beam travels through a dielectric material, it induces a microscopic displacement of charges, forming oscillating electric dipoles that combine to produce a macroscopic polarization. This leads to a change in the refractive index. When the electric field is intense enough, the response becomes nonlinear, giving rise to new propagation effects such as self-focusing, self-phase modulation, and plasma defocusing.

To derive the nonlinear polarization and the resulting Kerr effect in the SI system, we start from the definition of the electric displacement field:

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} \tag{11}$$

where $\mathbf{P}$ can be expanded to include the contribution of the third order susceptibility. Applied to a monochromatic electric field $\mathbf{E} = \mathcal{E} e^{i(k_0 z - \omega_0 t)} + \text{c.c.}$, this expression becomes:

$$\mathbf{P} = \varepsilon_0 \left(\chi^{(1)} \mathbf{E} + \chi^{(3)} \mathbf{E}^3 \right) \tag{12}$$

Expanding $\mathbf{E}^3$ and neglecting the $3\omega_0$ terms associated with third harmonic generation (THG), we isolate the term oscillating at the fundamental frequency, which represents the nonlinear response of the medium at the angular frequency ω_0:

$$\mathbf{P} \approx \varepsilon_0 \left[\chi^{(1)} + \frac{3}{4} \chi^{(3)} |\mathcal{E}|^2 \right] \mathbf{E} \tag{13}$$

Substituting this expression into the displacement field equation, we obtain:

$$\mathbf{D} = \varepsilon_0 \left(1 + \chi^{(1)} + \frac{3}{4} \chi^{(3)} |\mathcal{E}|^2 \right) \mathbf{E} = \varepsilon_0 \left(n_0^2 + \frac{3}{4} \chi^{(3)} |\mathcal{E}|^2 \right) \mathbf{E} = \varepsilon_0 \varepsilon_r \mathbf{E} \tag{14}$$

where $n_0^2 = 1 + \chi^{(1)}$. The effective refractive index n is defined as the square root of the relative permittivity, $n = \sqrt{\varepsilon_r} = \sqrt{n_0^2 + \frac{3}{4} \chi^{(3)} |\mathcal{E}|^2}$. For weak nonlinearities, a first order

Taylor expansion gives:

$$n \approx n_0 + \frac{3\chi^{(3)}}{8n_0}|\mathcal{E}|^2 \tag{15}$$

Comparing this expression with the standard form $n = n_0 + n_2 I$, where the intensity is $I = \frac{1}{2}n_0 c\varepsilon_0|\mathcal{E}|^2$, we identify the nonlinear refractive index coefficient:

$$n_2 = \frac{3\chi^{(3)}}{4n_0^2 c\varepsilon_0} \tag{16}$$

This refractive index therefore depends on intensity. This is the origin of the optical Kerr effect. Since the intensity $I(r)$ of a Gaussian laser beam is higher at the center than at the edges, the beam experiences a higher refractive index at the center than at the periphery, provided $\chi^{(3)} > 0$. This spatial variation of the refractive index acts exactly like a graded index lens, shown in Figure 4. The medium bends the wavefronts toward the high intensity axis. This phenomenon, known as self-focusing, is governed by the effective nonlinear polarization [4]:

$$\mathbf{P}_{\mathrm{NL}} = \frac{3}{4}\varepsilon_0\chi^{(3)}|\mathcal{E}|^2\mathbf{E} \tag{17}$$

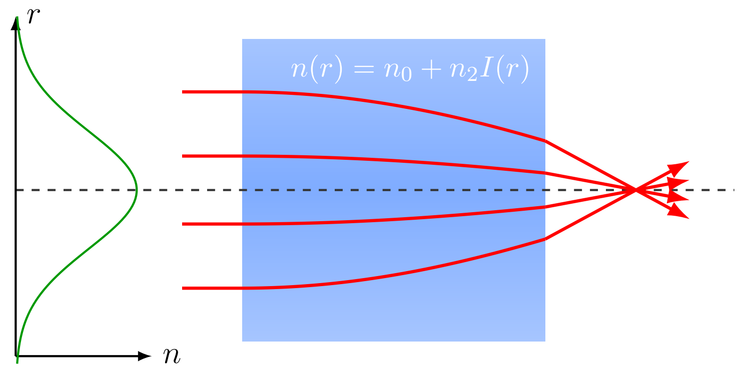


Figure 4: Kerr self-focusing as a graded index lens. The refractive index $n(r) = n_0 + n_2 I(r)$, plotted in green on the left, follows the transverse intensity profile of the beam, higher on axis where the beam is brightest (shaded blue gradient). Rays entering parallel to the axis are bent progressively more strongly as they approach the center, exactly as in a conventional graded index lens, and converge toward a nonlinear focus.

The phase of a femtosecond laser pulse in a dielectric depends on the nonlinear refractive index, $\Phi(t) = nk_0 z - \omega_0 t = n_0 k_0 z + n_2 k_0 z I - \omega_0 t$. Differentiating with respect to time gives the instantaneous frequency shift due to self-phase modulation:

$$\frac{d\Phi}{dt} = -\omega_0 + n_2 z\frac{\omega_0}{c}\frac{dI(t)}{dt} \tag{18}$$

On the rising edge of the pulse, $dI/dt > 0$ and the instantaneous frequency is shifted toward the red, while on the falling edge, $dI/dt < 0$ and it is shifted toward the blue.

This self-phase modulation alone, within the slowly varying envelope approximation, does not distort the temporal profile of the pulse. It only adds a frequency drift without changing the shape of $I(t)$.

Self-steepening proper is a distinct effect, carried by the shock term $(2n_2/c)\,\partial(|\mathcal{E}|^2\mathcal{E})/\partial t$, which comes from the fact that the group velocity itself depends on intensity through $n_2 I$ [12]. The more intense peak of the pulse therefore travels more slowly than the wings, which progressively pulls the intensity maximum toward the trailing edge and steepens this edge more and more with propagation distance [7]. Contrary to what a quick reading of the self-phase modulation equation alone would suggest, this shock term distorts the temporal profile in an intrinsically asymmetric way, even for a perfectly symmetric Gaussian input pulse, and it is identified as responsible for the asymmetry observed in supercontinuum generation spectra [12, 8].

2.5 Plasma Defocusing

Several effects generate electrons in the conduction band. These include thermal excitation, tunnel ionization, multiphoton ionization, and free electrons resulting from crystal defects. All these free carriers respond to the incident electric field. The simplest way to model this is to treat the conduction electrons as a free electron gas subject to friction $\gamma = 1/\tau$. The equation of motion in a field $\vec{E}(t) = \vec{E}_0\, e^{-i\omega_0 t}$ reads:

$$m^* \frac{d^2}{dt^2}\vec{x} = -e\vec{E} - \frac{m^*}{\tau}\frac{d}{dt}\vec{x} \implies \vec{x} = \frac{\tau e\vec{E}}{m^*\omega_0\left(\omega_0\tau + i\right)} \tag{19}$$

The polarization of the free carriers $\vec{P}_{\text{free}} = -eN\vec{x}$ follows:

$$\vec{P}_{\text{free}} = -eN\vec{x} = -\frac{\omega_p^2\tau}{\omega_0\left(\omega_0\tau + i\right)}\varepsilon_0\vec{E} \tag{20}$$

where we introduce the plasma frequency $\omega_p^2 = \dfrac{Ne^2}{\varepsilon_0 m^*}$.

The valence electrons are not removed by the excitation, and their polarization $\vec{P}_{\text{bound}} = \varepsilon_0\chi^{(1)}\vec{E}$ must be conserved. Both populations are driven by the same macroscopic field and respond linearly, so their contributions add up. Introducing the linear index $n_0^2 = 1 + \chi^{(1)}$ of the unexcited medium, the relative permittivity reads:

$$\varepsilon_r = \frac{D}{\varepsilon_0 E} = \frac{1}{\varepsilon_0\vec{E}}\left(\varepsilon_0\vec{E} + \vec{P}_{\text{bound}} + \vec{P}_{\text{free}}\right) = n_0^2 - \frac{\omega_p^2\tau}{\omega_0(\omega_0\tau + i)} \tag{21}$$

Multiplying numerator and denominator by the complex conjugate and separating the real and imaginary parts, we obtain:

$$\varepsilon_r = \varepsilon_1 + i\varepsilon_2 \tag{22}$$

$$\varepsilon_1 = n_0^2 - \frac{\omega_p^2\tau^2}{1 + \omega_0^2\tau^2}, \tag{23}$$

$$\varepsilon_2 = \frac{\omega_p^2\tau}{\omega_0\left(1 + \omega_0^2\tau^2\right)} > 0, \tag{24}$$

The complex index follows from $\tilde{n} = n + i\kappa = \sqrt{\varepsilon_r}$, so $\varepsilon_1 = n^2 - \kappa^2$ and $\varepsilon_2 = 2n\kappa$.

When the number of oscillations before a collision is large, the real part of the refractive index simplifies to:

$$n = \sqrt{\varepsilon_1} = \left(n_0^2 - \frac{\omega_p^2\tau^2}{1 + \omega_0^2\tau^2}\right)^{\frac{1}{2}} \approx n_0\left(1 - \frac{\omega_p^2}{n_0^2\,\omega_0^2}\right)^{\frac{1}{2}} \tag{25}$$

The square root must be real, which requires $\dfrac{\omega_p^2}{n_0^2\,\omega_0^2} \leq 1$. Expanding the definition of the plasma frequency, we obtain:

$$\frac{Ne^2}{n_0^2\,\varepsilon_0 m^*\omega_0^2} \leq 1 \implies N \leq \frac{n_0^2\,\varepsilon_0 m^*\omega_0^2}{e^2} \tag{26}$$

Defining the critical density N_c as the density at which the plasma frequency equals the laser frequency, $\omega_p(N_c) = \omega_0$, we have:

$$N_c = \frac{\varepsilon_0 m^* \omega_0^2}{e^2} \implies \frac{\omega_p^2}{\omega_0^2} = \frac{N}{N_c} \tag{27}$$

The condition then becomes $N \leq n_0^2 N_c$, beyond which the wave is evanescent and the plasma becomes reflective. The index is then:

$$n = n_0 \sqrt{1 - \frac{\omega_p^2}{n_0^2\,\omega_0^2}} = n_0 \sqrt{1 - \frac{N}{n_0^2 N_c}} \tag{28}$$

For an underdense plasma, $N \ll n_0^2 N_c$, the real part of the refractive index can be expanded as:

$$n \simeq n_0 - \frac{N}{2\,n_0 N_c} \tag{29}$$

This expression and the definition of the critical density N_c match the Drude model as it is used to describe beam defocusing by the free electron plasma generated by the pulse [4].

2.6 Filamentation

The propagation of femtosecond laser pulses in transparent dielectrics such as SiO_2 can lead to the formation of filaments. This self guided mecanism originate from a dynamic balance between Kerr self-focusing and plasma defocusing. This phenomenon occurs when the input power exceeds a critical value, denoted P_{cr}, is illustrated in Figure 5.

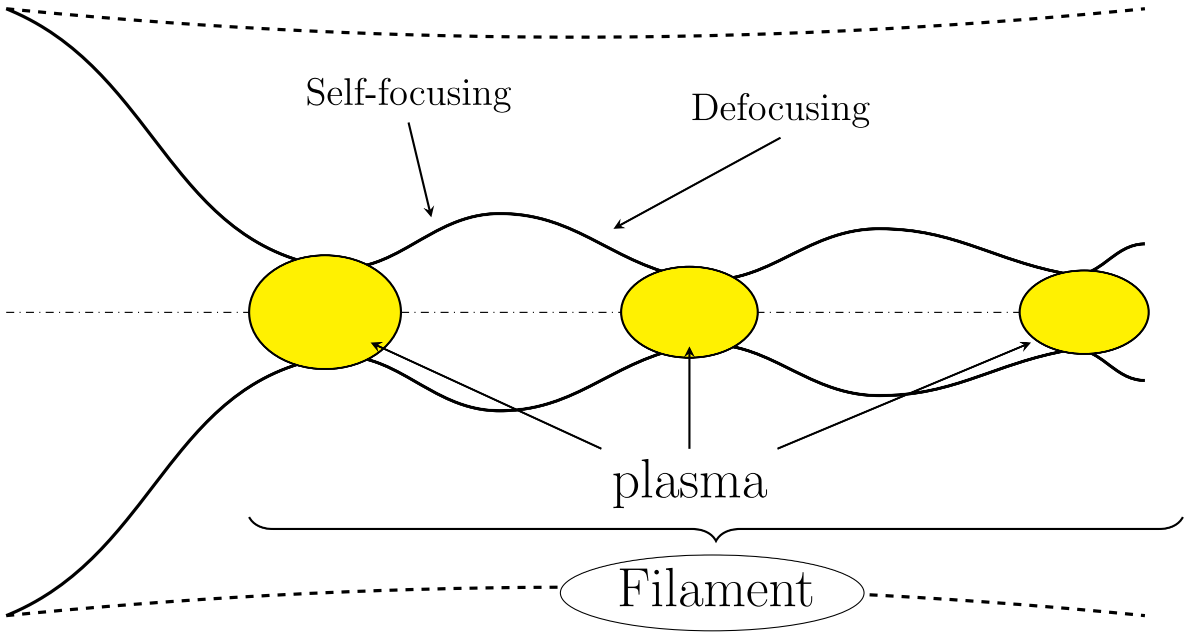


Figure 5: Schematic beam envelope undergoing repeated cycles of Kerr self-focusing and plasma defocusing. Each collapse is arrested before catastrophic self-focusing by the free electron plasma generated near the intensity peak (yellow), which defocuses the beam until the intensity drops enough for the plasma to recombine and self-focusing to take over again, refilled by the surrounding energy reservoir. This repeated breathing over an extended propagation distance is the filament.

The critical power P_{cr} follows from the balance between the nonlinear self-focusing effect and diffraction. For a Gaussian beam, it is given by [10]:

$$P_{cr} = \frac{3.77\lambda_0^2}{8\pi n_0 n_2}, \tag{30}$$

where λ_0 is the laser wavelength, n_0 the linear refractive index, and n_2 the nonlinear index coefficient. The numerical factor is 3.77 for a Gaussian beam and 3.72 for the Townes profile, for which diffraction and self-focusing balance exactly.

Above threshold, the distance at which an unopposed beam would collapse is given by the semi-empirical formula fitted to numerical solutions of the nonlinear Schrödinger equation by Dawes and Marburger [11, 10]:

$$L_c = \frac{0.367\, L_{DF}}{\sqrt{\left[(P_{in}/P_{cr})^{1/2} - 0.852\right]^2 - 0.0219}}, \tag{31}$$

where $L_{DF} = k_0 w_0^2/2$ is the Rayleigh length. For an externally focused beam of focal length f, the collapse distance shortens according to $L_{c,f}^{-1} = L_c^{-1} + f^{-1}$. Equation (31) is what sets the position of the nonlinear focus in the simulations, before plasma defocusing arrests the collapse.

For fused silica, typical values at $\lambda_0 = 800$ nm are $n_0 \approx 1.45$ and $n_2 \approx 2.4 \times 10^{-20}$ m^2/W, a value measured by Taylor et al. and reported in the reference compilation of Milam [13]. Using these values, we calculate $P_{cr} \approx 2.8$ MW, consistent with the 1.98 to 2.5 MW reported respectively in [6] and [9] for fused silica at this wavelength. At $\lambda_0 = 1030$ nm, since no measurement exists exactly at this wavelength, we use the closest value available in the literature, $n_2 \approx 2.74 \times 10^{-20}$ m^2/W at 1053 nm [13], which gives $P_{cr} \approx 4.0$ MW.

The critical power is related to the laser intensity I through the beam cross section. For a Gaussian beam of radius w_0, the peak intensity is:

$$I_0 = \frac{2P}{\pi w_0^2}. \tag{32}$$

In the tight focusing experiments of [6], the measured beam radius at focus ranges from 0.7 to 4.8 μm depending on the objective used, with $w_0 = 1.1$ μm for the 20×, NA = 0.5 objective retained in that reference. Taking this radius with $P = P_{cr} \approx 2.8$ MW, we obtain an intensity of order 1.5×10^{14} W/cm^2, higher than the clamping intensity of about 5×10^{13} W/cm^2 actually measured in [6]. This gap is expected. $I_0 = 2P/(\pi w_0^2)$ assumes linear focusing all the way down to the radius w_0, whereas in reality the defocusing produced by the plasma generated during propagation limits the peak intensity well before the beam reaches this radius, which is why the observed clamping intensity remains lower than this simple geometric estimate.

Note for revision. Once `notebooks/term_ablation_study.ipynb` from `deliverable (3)` has actually been run, this is where the resulting figures belong. Specifically: peak intensity versus z (reproducing Fig. 8 of [6]) and the fluence contour maps with the FWHM/2 curve versus z (reproducing Fig. 7) both belong right here, at the end of this subsection, since they directly test the P_{cr} and clamping intensity claims just discussed. The on-axis electron density versus z (Fig. 9) fits better at the end of Section 11.2 (Carrier Density Rate Equations), since it is a direct check of the rate equations rather than of the beam envelope. The comparison of free versus trapped carriers and the pulse intensity at fixed z (Fig. 13) fits best in Section 10, next to the self-trapped exciton discussion. Send me the generated PNGs or the notebook output once you have them and I will place them at these exact spots with proper captions.

2.7 Exciton Formation

When an electron is promoted from the valence band to the conduction band, for example by optical absorption, it leaves behind a hole of effective charge $+e$. Electrons in the conduction band can feel the potential created by the holes in the valence band. This electrostatic attraction can be modeled by a Coulomb potential, which allows us to solve the Schrödinger equation for a composite particle called an exciton, made of a conduction band electron and a valence band hole.

This quasiparticle has a reduced mass defined by $\frac{1}{\mu} = \frac{1}{m_e} + \frac{1}{m_h}$ and a quantized energy spectrum of the hydrogen-like type.

In the absence of Coulomb interaction, the minimum energy required to create a free electron-hole pair corresponds to the gap energy E_g. However, because of the attractive Coulomb potential between the electron and the hole, they can form a bound state. The accessible energy levels then sit below the conduction band. These states can be populated directly through the process described above, but also through nonradiative relaxation mechanisms such as the Auger effect or exciton-phonon interactions. As pointed out by Mao et al. [4], this formation process is very fast, often under 1 ps for wide bandgap materials.

The relative motion of this electron-hole pair can be modeled by a hydrogen-like Schrödinger equation. The stationary equation for the relative coordinate $\mathbf{r}$ reads:

$$\left[\frac{\hbar^2}{2\mu}\nabla^2 + \frac{e^2}{4\pi\epsilon_0\epsilon_r r}\right]\psi(\mathbf{r}) = E_b\psi(\mathbf{r}), \tag{33}$$

where $\epsilon = \epsilon_0\epsilon_r$ is the dielectric permittivity of the medium and $E_b > 0$ is the exciton binding energy.

The relation between the gap energy E_g, the binding energy term, and the excitation energy E_n required for the n-th exciton state is given by:

$$E_g = E_n + \frac{R_\chi}{n^2}, \tag{34}$$

where R_χ is the effective exciton Rydberg energy:

$$R_\chi = \frac{\mu e^4}{2(4\pi\epsilon)^2\hbar^2} = R_H\left(\frac{\mu}{m_0}\right)\frac{1}{\epsilon_r^2}. \tag{35}$$

Here, $R_H \approx 13.6$ eV is the Rydberg constant of the hydrogen atom, m_0 is the free electron mass, and ϵ_r is the relative dielectric constant of the material.

To evaluate the spatial extent of the exciton wavefunction, we define the exciton Bohr radius a_χ:

$$a_\chi = \frac{4\pi\epsilon\hbar^2}{\mu e^2} = a_0 \cdot \epsilon_r\left(\frac{m_0}{\mu}\right), \tag{36}$$

where $a_0 \approx 0.53$ Å is the atomic Bohr radius.

In wide bandgap materials, electrons are strongly bound to their atoms, which means the relative dielectric permittivity is low for SiO_2 or alkali halides, where the relative dielectric constant is typically of order 2 to 4. As a result, the material provides weak dielectric screening of the Coulomb attraction, maintaining a deep potential well.

A low permittivity directly leads to a reduction of the exciton Bohr radius, down to a few angstroms. When a_χ becomes comparable to the interatomic spacing of the lattice, the exciton localization no longer extends over several lattice atoms but shifts to a strongly bound regime, illustrated in Figure 6. The binding energy also increases, preventing thermal dissociation and favoring ultrafast energy relaxation. During this process, the excess electrostatic potential energy of the unscreened exciton is abruptly discharged into local lattice vibrations (phonons) through massive exciton-phonon coupling within a few picoseconds. This induces a local lattice distortion and leads to the formation of a self-trapped exciton (STE).

(a) Weakly bound exciton, large permittivity ϵ_r: the wavefunction envelope extends over many lattice sites, close to a free electron-hole pair.

(b) Strongly bound exciton, low permittivity: the wavefunction collapses onto a single lattice site, the regime relevant for SiO_2.

Figure 6: Spatial extent of the exciton wavefunction relative to the lattice spacing. The grey dots represent lattice atoms along one direction, the colored curve the envelope $|\psi(\mathbf{r})|$ of the relative electron-hole wavefunction. A smaller ϵ_r shrinks a_χ from Eq. (36) until it reaches the strongly bound regime on the right.

2.8 Self-Trapped Excitons

Thermal fluctuations induce slight lattice vibrations that act as a trigger, creating a transient local deformation. The strongly localized exciton is captured by this distortion. Through massive exciton-phonon coupling, it abruptly discharges its potential energy into the lattice, considerably deepening the local potential well, shown schematically in Figure 7.

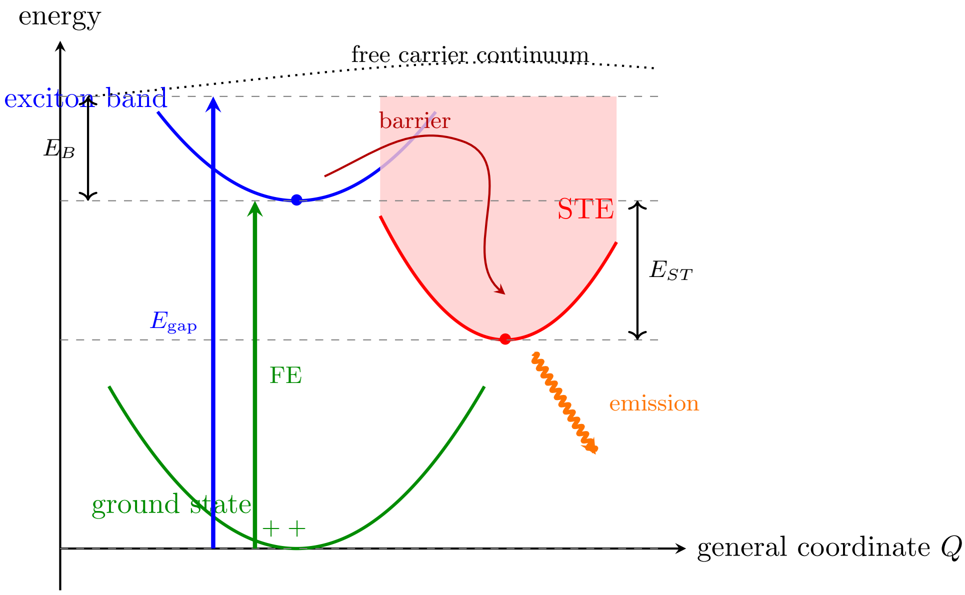
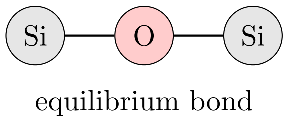
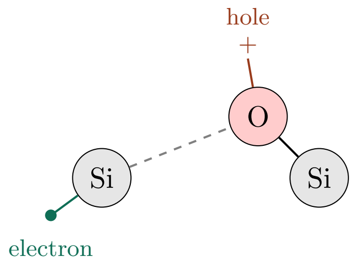


Figure 7: Configuration coordinate diagram for self-trapping, adapted to the general shape reported for oxide self-trapped excitons. Absorption near E_{gap} populates the free carrier continuum, which relaxes into the exciton band, bound below the continuum by E_B. In materials with weak exciton-phonon coupling a barrier separates the exciton band from a deeper, displaced self-trapped exciton (STE) well, of depth E_{ST} relative to the exciton band. In SiO_2 the coupling is strong enough that this barrier is negligible, so self-trapping proceeds essentially without delay once the exciton is formed, as described in the text. The STE eventually relaxes back to the ground state, releasing its energy as an emission line strongly redshifted from E_{gap} and, in SiO_2, leaving behind the permanent bond rearrangement of Figure 8.

This intense localized energy weakens and breaks the Si–O–Si covalent bond, displacing the oxygen atom from its equilibrium position. This structural rupture generates two dangling bonds, each holding an unpaired electron. In this strongly distorted configuration, the exciton self-traps. The hole localizes on the oxygen dangling bond, while the electron remains bound to the silicon dangling bond [4], as sketched in Figure 8.



(a) Before: intact Si–O–Si bridge, oxygen at its equilibrium position.



(b) After: the Si–O bond breaks, the oxygen relaxes off site, leaving a silicon dangling bond carrying the electron and an oxygen dangling bond carrying the hole.

Figure 8: Schematic bond-breaking mechanism of self-trapping in SiO_2, simplified to a linear Si–O–Si fragment for clarity (the real network is tetrahedrally coordinated). The two dangling bonds left behind are the microscopic identity of the self-trapped exciton.

This mechanism explains the formation of Self-Trapped Excitons (STE) in silica, where the excitation energy is rapidly converted into local lattice distortion, opening the way to permanent structural modifications of the material [4].

3 Propagation Equation and Population Rate Equations

Before describing how the simulation code is built, we assemble the full envelope propagation equation and the carrier density rate equations exactly as they are implemented in `NewSim3juillet.py`. The physical building blocks were derived separately in the previous sections. Kerr self-focusing and self-steepening in Section 6, plasma defocusing and absorption in Section 7, photoionization in Section 3, and avalanche and self-trapping in Section 5. Here we put them together into the single equation that the code actually integrates, and we label every term so that the correspondence with the source code is explicit.

3.1 Slowly Varying Envelope Equation

Following the same derivation as in Section 6, the electric field is written as a carrier wave modulated by a slowly varying envelope $u(r, t, z)$, and the wave equation is reduced under the slowly varying envelope approximation. For fused silica, the resulting propagation equation used in the code reads:

$$\hat{U}\,\frac{\partial u}{\partial z} = \underbrace{\frac{i}{2k_0}\nabla_\perp^2 u}_{\text{diffraction}} + \underbrace{i\,\hat{D}\,\hat{U}\,u}_{\substack{\text{dispersion: exact Sellmeier, all orders}\\ \text{leading term } -\frac{i}{2}k_\omega''\,\partial_t^2 u}}$$

$$+ \underbrace{i\,\hat{T}^2\,\frac{3\omega_0^2\chi^{(3)}}{8k_0c^2}(1-f_R)\,|u|^2\,u}_{\text{instantaneous Kerr effect and self-steepening}}$$

$$+ \underbrace{i\,\hat{T}^2\,\frac{3\omega_0^2\chi^{(3)}}{8k_0c^2}\,f_R\left(R * |u|^2\right)u}_{\text{delayed Raman response}}$$

$$- \underbrace{\hat{T}\,\frac{U_i\,W_{PI}(|u|)}{n_0c\varepsilon_0|u|^2}\left(1-\frac{N}{N_{at}}\right)u}_{\text{photoionization energy loss}} - \underbrace{\frac{\sigma_\omega}{2}(1+i\omega_0\tau_c)N\,u}_{\text{plasma absorption and defocusing}}$$

$$+ \underbrace{i\,\frac{\omega_0}{2n_0c\,\rho_c}\frac{\omega_0^2}{\omega_{tr}^2-\omega_0^2}\,N_{STE}\,u}_{\text{bound STE polarizability, new in Section 11.4}} \tag{37}$$

with, in the retarded frame $t \to t - z/v_g$ and writing $\Omega = \omega - \omega_0$,

$$\hat{T} = 1 + \frac{i}{\omega_0}\frac{\partial}{\partial t} \longleftrightarrow 1 + \frac{\Omega}{\omega_0}, \qquad \hat{U} = 1 + \frac{ik_\omega'}{k_0}\frac{\partial}{\partial t} \longleftrightarrow 1 + \frac{k_\omega'\Omega}{k_0}, \tag{38}$$

$$\hat{D} \longleftrightarrow k(\omega) - k_0 - k_\omega'\Omega, \tag{39}$$

where $k(\omega)$ is evaluated from the Sellmeier relation rather than truncated at any order.

Three points in this equation deserve to be stated explicitly, because each of them is a place where a plausible-looking alternative would be wrong.

First, $\hat{U}$ sits in front of $\partial/\partial z$, and the transverse Laplacian is bare. This is the form used by Couairon et al. [6], whose Eq. (2) is introduced with exactly that wording: the equation accounts for space-time focusing *due to the presence of the operator $\hat{U}$ in front of $\partial/\partial z$*. The reason to prefer it over the algebraically equivalent form obtained by dividing through by $\hat{U}$ is that it contains no inverse operator, so there is no ambiguity about which terms $\hat{U}$ acts on. If one does divide by $\hat{U}$, it must be applied to *every* term on the right-hand side, giving

$$\frac{\partial u}{\partial z} = \frac{i}{2k_0}\hat{U}^{-1}\nabla_\perp^2 u + i\hat{D}u + \hat{U}^{-1}\Big\{\text{Kerr} + \text{Raman} + \text{ionization} + \text{plasma} + \text{STE}\Big\}, \tag{40}$$

where $\hat{U}$ cancels on the dispersion term alone. Since $\hat{U} \simeq \hat{T}$ to within one percent for fused silica at 800 nm ($n_g/n_0 = 1.0095$), the Kerr operator in that form becomes $\hat{U}^{-1}\hat{T}^2 \simeq \hat{T}$ and the ionization operator becomes $\hat{U}^{-1}\hat{T} \simeq 1$, which is how the single-$\hat{T}$ expressions of the review literature arise. Applying $\hat{U}^{-1}$ to the Laplacian only, while leaving $\hat{T}^2$ on the Kerr term, is *not* one of these two forms and leaves every nonlinear term with one power of $\hat{T}$ too many.

Second, the depletion factor is the dimensionless $1 - N/N_{at}$, not $N_{at} - N$. The Keldysh rate W_{PI} returned by the lookup table is already a volumetric rate in $\text{cm}^{-3}\text{s}^{-1}$, so multiplying it by a density would leave the term with an extra cm^{-3}. The alternative convention, in which W_{PI} is a per-atom rate in s^{-1} multiplied by $N_{at} - N$, is equally valid but is not the quantity the code computes.

Third, the group delay $k_\omega'\Omega$ does not appear as a separate term: it has been absorbed into $\hat{D}$, because the code works in the retarded frame and subtracts it inside `delta_k`. Writing it again outside would count it twice.

How the solver implements this. Equation (37) cannot be stepped forward as written, since it gives $\hat{U}\partial_z u$ rather than $\partial_z u$; the code therefore integrates the divided form (40). The linear half step supplies the first two terms, applying `inv_U_op` to the Laplacian and leaving `delta_k` bare, and the nonlinear step assembles the whole brace in frequency space and multiplies it by $\hat{U}^{-1}$ in one operation, so that the Kerr, Raman, ionization, plasma and STE terms all receive it. Because $\hat{U}^{-1}$ here multiplies source terms directly rather than sitting inside a phase factor $e^{i\varphi}$, it is floored and masked before use: $\hat{U}$ vanishes at $\Omega = -k_\omega'^{-1} k_0 \simeq -0.99\,\omega_0$, that is at essentially zero absolute frequency. The floor only acts beyond $\lambda = 7.4\ \mu$m, outside the Sellmeier window where the spectral mask has already suppressed the field, so within the physical band $\hat{U}^{-1}$ stays between 0.22 and 6.6 and is not distorted.

3.2 Where the Self-Steepening Operator Comes From, and Why It Is Squared on Some Terms

Section 6 introduced self-steepening through the time domain shock term $(2n_2/c)\,\partial(|\mathcal{E}|^2\mathcal{E})/\partial t$. The propagation equation above instead writes it as a frequency domain multiplication by $T = 1 + \Omega/\omega_0$, or T^2 on some terms. These are the same physics in two different languages, and the reason some terms get T, some get T^2, and the plasma term gets no such operator at all, comes directly from how many time derivatives that term carries before the slowly varying envelope approximation is applied.

Every source term on the right hand side of the underlying wave equation originates from a current or a polarization evaluated at the true, unapproximated frequency $\omega = \omega_0 + \Omega$ of each spectral component, not at the fixed carrier frequency ω_0. Writing $\omega = \omega_0(1 + \Omega/\omega_0) = \omega_0 T$, any factor of ω appearing in a source term becomes a factor of $\omega_0 T$ once it is pulled out and referenced to the carrier frequency, exactly as $k(\omega)$ was expanded around k_0 for the dispersion term. The Kerr polarization enters the wave equation through $\partial^2 P_{NL}/\partial t^2$, a second time derivative, which in the frequency domain is a factor of $-\omega^2 = -\omega_0^2 T^2$. The photoionization current, like the plasma current, enters through a single time derivative $\partial J/\partial t$, a factor of $-i\omega = -i\omega_0 T$. This is the entire reason the Kerr and Raman terms above carry T^2 while the photoionization loss term carries only T. Both are the same operator $T = 1 + \Omega/\omega_0$, applied once for every power of ω that the term's own derivation contributes, and Section 11.4 of `deliverable (3)` corrects the July code precisely because it had applied T to the Kerr term as if it only carried one such derivative.

The plasma current J_{plasma} also comes from a single time derivative, so by the same counting it should carry T^1 as well. It appears without any T in the equation above because the Drude derivation of Section 7 already keeps the full frequency dependence of $\sigma(\omega)$ and $(1 + i\omega\tau_c)$ exactly, rather than expanding around ω_0 first, so the correction is already built into σ_ω itself and does not need to be reintroduced as a separate multiplicative operator.

Physically, T being larger than one on the blue side of the spectrum ($\Omega > 0$) and smaller than one on the red side is exactly the same statement as the shock term of Section 6 pulling energy toward the trailing edge of the pulse: whichever nonlinear term this operator multiplies, it systematically favors the blue shifted, trailing part of the spectrum over the red shifted, leading part, which is the frequency domain signature of the group velocity depending on intensity. Squaring the operator on the Kerr term simply means this asymmetry is applied twice, once for each power of ω that the second time derivative of P_{NL} contributes, and is why the Kerr induced spectral asymmetry is normally the dominant contribution to self-steepening compared to the single power of T on photoionization loss.

Each underbraced term maps onto a specific piece of the code. Diffraction and dispersion are applied together by `LinearOperator.half_linear`, in the quasi discrete Hankel basis described in Section 12. They are combined into a single step rather than split into two because both are diagonal once expressed in (Hankel radial mode, temporal frequency Ω) space, so they commute and applying them together is exact, at the cost of one FFT/IFFT pair per call instead of two. The space-time focusing factor enters this same step as `inv_U_op` $= 1/\hat{U}(\Omega)$ multiplying the transverse Laplacian, and reduces to the identity when `enable_space_time_focusing` is off. Rather than truncating the Taylor expansion of $k(\omega)$ at second order as we did analytically in the derivation of Section 6, the code evaluates $k(\omega)$ directly from the Sellmeier relation for every frequency component and subtracts the linear term $k_0 + k'_\omega\Omega$, so the operator `delta_k` used in the code already contains dispersion to all orders, not just group velocity dispersion.

The Kerr, self-steepening, and Raman terms are computed together in `NonlinearOperator.split`. The instantaneous part $(1 - f_R)|u|^2$ and the delayed Raman part $f_R\,(R * |u|^2)$, obtained by convolving the intensity with the retarded Raman response function $R(t)$, are summed before multiplying by u, which is why they are written here as two separate lines rather than as two independent physical effects. The self-steepening operator $\left(1 + \frac{i}{\omega_0}\partial_t\right)$ is applied in the frequency domain as `T_op` $= 1 + \Omega/\omega_0$, following the same operator introduced by Mao et al. [4] for free carrier absorption and used identically here for the Kerr response.

The photoionization loss term removes energy from the field at exactly the rate needed to promote electrons to the conduction band at the Keldysh rate W_{PI}, throttled by the fraction $1 - N/N_{at}$ of ionizable sites still neutral, consistent with the ionization loss derivation of Section 11.3 below. N_{at} is the total density of ionizable sites (`rho_max` in the code) and N is the current free electron density; in the code this factor is `depl_field = clip(1 - rho/rho_max, 0, 1)`. The plasma absorption and plasma induced defocusing are grouped into a single term above since the code computes them together as the real and imaginary parts of the same Drude response derived in Section 7, with σ_ω the inverse Bremsstrahlung cross section already introduced in Section 5.

Note for revision. The Raman response function $R(t)$ used in the code, with time constants `tau_s` and `tau_d`, is the standard damped oscillator form used for silica nonlinear optics. If you want the explicit derivation of $R(t)$ from a driven harmonic oscillator model included here, let me know and I will add it with the matching reference.

Update relative to the July solver (`deliverable (3)`). Eq. (4) of Couairon et al. [6] squares the self-steepening operator on the Kerr and Raman terms, $\left(1 + \frac{i}{\omega_0}\partial_t\right)^2$, applies it only to the first power on the photoionization loss term, and applies none at all to the plasma term. The July code in `NewSim3juillet.py` used a single `T_op` on both the Kerr and the photoionization terms, and `deliverable (3)` initially changed this to `T_op**2` on the Kerr term to match the paper.

That change, taken alone, was a regression, and it is worth recording why because the reasoning looks sound. Couairon's $\hat{T}^2$ lives in his Eq. (2), which carries $\hat{U}$ in front of $\partial/\partial z$; the July code had no $\hat{U}$ anywhere, so its single $\hat{T}$ was already the correct value of $\hat{U}^{-1}\hat{T}^2$ for a model without space-time focusing. Importing the numerator of one form while keeping the left-hand side of the other left the self-steepening roughly twice too strong: expanding in $\varepsilon = \Omega/\omega_0$ gives $\hat{T}^2 = 1 + 2\varepsilon + \varepsilon^2$ against the correct $\hat{U}^{-1}\hat{T}^2 = 1 + 0.99\,\varepsilon$, and since self-steepening *is* the departure from unity, the modelled effect was doubled at leading order, rising to a factor of three on the far blue wing. The present version resolves this by applying $\hat{U}^{-1}$ to the entire nonlinear bracket, as described above, which reproduces Eq. (2) exactly rather than approximately and, unlike the shortcut of simply reverting the Kerr term to $\hat{T}^1$, also supplies the $\hat{U}^{-1}$ that the plasma and STE terms require. The solver keeps the spectral mask `spec_mask` multiplying `T_op`, since $T = 1 + \Omega/\omega_0$ becomes negative for absolute frequencies below zero and this sign flip was found to be numerically unstable, amplified by up to a factor 9 per step once the operator is squared. The mask is a smooth tanh window that suppresses the field outside the wavelength range where the Sellmeier fit is valid,

$\lambda \in [0.18, 5]$ μm, and is controlled by `enable_spectral_filter`, default on. It has no effect on the physical, non-negative-frequency part of the spectrum. Six new boolean switches, one per term of the equation above (instantaneous Kerr, Raman, self-steepening, photoionization loss, plasma defocusing, plasma absorption, named `enable_kerr_instantaneous` through `enable_plasma_absorption` in the code), were also added so each term can be switched off independently. This is what the ablation study of the accompanying notebook uses to test what every term actually does to the beam rather than only reading it from the equation.

3.3 Carrier Density Rate Equations

The code integrates two coupled populations, the free electron density N and the self-trapped exciton density N_{STE}, exactly as introduced in Section 5:

$$\begin{aligned}\frac{dN}{dt} = & \underbrace{W_{PI}(|u|)\,(N_{at}-N)}_{\text{direct photoionization, depletes available sites}} \\ & + \underbrace{\Big[W_{PI}(|u_x|) + \beta_s I\,N\Big]\frac{N_{STE}}{N_{at}}}_{\text{re-ionization of trapped excitons}} \\ & + \underbrace{\frac{\sigma_\omega}{U_i} I\,N\,(1-N/N_{at})}_{\text{avalanche ionization}} - \underbrace{\frac{N}{\tau_r}}_{\text{trapping into the STE channel}}\end{aligned} \tag{41}$$

$$\frac{dN_{STE}}{dt} = \underbrace{\frac{N}{\tau_r}}_{\text{trapping}} - \underbrace{\Big[W_{PI}(|u_x|) + \beta_s I\,N\Big]\frac{N_{STE}}{N_{at}}}_{\text{re-ionization}} \tag{42}$$

where $I = n_0 c \varepsilon_0 |u|^2/2$ is the local intensity, u_x is the field evaluated with the smaller gap U_s of the trapped state instead of U_i, so that $W_{PI}(|u_x|)$ is the Keldysh rate computed for the shallower STE gap, and β_s is the avalanche coefficient for the STE channel. This two population model with a distinct gap U_s for the trapped state follows the parametrization used in the code (`Us_eV`, labeled Chimier PRB 2011 in the source comments).

Note for revision. I have not been able to check this rate equation against the original Chimier, PRB 84, 075092 (2011) paper, since it is not among the sources currently in the repository. If you upload it, I will verify the coefficients β_s and U_s against it directly and correct this section if needed.

Note the correspondence with Section 5. There, we wrote the same pair of equations with σ/E_g in place of σ_ω/U_i and τ_x in place of τ_r. These are the same quantities, only the notation has been aligned here with the variable names used in the code (`tau_r` for the trapping time, `avalanche_coef` for σ_ω/U_i).

Update relative to the July solver. The July code only let the self-trapped exciton population decay by re-ionization back into the conduction band, the $\big[W_{PI}(|u_x|)+\beta_s IN\big]N_{STE}/N_{at}$ term above. The parametrization referenced in the code (Sakurai et al.) tabulates for fused silica both a trapping time of 150 fs, already present as τ_r, and a separate relaxation time of the trapped state itself of about 1 ps, which was missing. `deliverable (3)` adds an optional term $-N_{STE}/\tau_{STE}$ to the dN_{STE}/dt equation, controlled by `Config.tau_ste`, defaulting to `None` which reproduces the July behavior exactly. This mainly matters for the probe phase at picosecond delays, since it changes how long the STE population, and therefore the index change it produces (Section 11.4), survives after the pump pulse has left.

3.4 Photoionization Energy Loss

For completeness we derive the photoionization loss term used above, since it does not appear explicitly in the earlier sections. Each ionization event removes one photoelectron from the neutral population and costs at least the gap energy U_i, which is taken directly from the field. The absorbed power per unit volume is:

$$\mathcal{P}_{ion} = U_i \, (N_{at} - N) \, W_{PI}(|u|) \tag{43}$$

Writing this as a loss term $-\alpha_{ion} I$ acting on the intensity $I = n_0 c \varepsilon_0 |u|^2/2$ and solving for α_{ion} gives the coefficient used above:

$$\alpha_{ion} = \frac{U_i \, (N_{at} - N) \, W_{PI}(|u|)}{n_0 c \varepsilon_0 |u|^2/2} \tag{44}$$

which is exactly the `photo` variable computed inside `NonlinearOperator.split`.

3.5 Bound Polarizability of Self-Trapped Excitons

This subsection documents a term that did not exist at all in the July solver and is new in `deliverable (3)`. A self-trapped exciton is not a conduction band carrier. It is a pair of dangling bonds, $Si^{\bullet}$ and $O^{\bullet}$, left behind after the Si-O-Si bond breaks, described in Section 10, and its unpaired orbitals sit inside the gap rather than in the conduction band. It therefore does not drift in the field and does not contribute to the free carrier Drude response of Section 7. What it does instead is polarize as a bound oscillator with a resonance at the energy of its first excited level, following Mao et al. [4]:

$$\Delta n_{STE} = \frac{N_{STE} \, e^2}{2 n_0 m_e \varepsilon_0} \frac{1}{\omega_{tr}^2 - \omega^2} \tag{45}$$

The July code, and the interferometry section of this report, only ever evaluated this formula for the probe beam in post-processing, in `unified_filament_slider_v3.py`. It never entered the propagation equation itself, so the pump pulse never felt the index change its own trapped population produces. `deliverable (3)` adds this contribution directly to the field equation as the fifth line above, controlled by `enable_ste_index`, default on, and implemented as a pure phase term with no associated loss, since the pump photon energy at 1.55 eV is far from the STE resonance, so there is no one-photon absorption to account for on the pump itself. Because the pump lies below the resonance, $\omega_0 < \omega_{tr}$, this term is positive and adds to the Kerr self-focusing rather than opposing it, which is also the mechanism identified as the permanent, fracture-free index change that Couairon et al. classify as type I damage.

A second correction accompanies this addition. The resonance energy itself was taken as 5.2 eV in the July code, but Mao et al. specify 5.2 eV as the peak of the STE transient absorption band, not the resonance frequency to use for the refractive index formula above, for which they give the first excited level at $\omega_{tr} \approx 4.2$ eV. `deliverable (3)` corrects `E_tr_eV` to 4.2, which changes the resonance denominator at the 490 nm probe wavelength used in this report by a factor of about 1.84.

Note for revision. I have not independently re-derived the bound oscillator formula for Δn_{STE} from first principles here, I have only checked that the code matches the Mao et al. expression term for term. If you want the Lorentz oscillator derivation included for completeness, let me know.

4 Numerical Method

The propagation equation of Section 11 mixes a transverse diffraction term, a temporal dispersion term, and a nonlinear term that depends locally on the field at every point in space and time. No single numerical scheme handles all three cheaply. The code therefore treats each physical mechanism with the tool best suited to it, and recombines them at every propagation step through operator splitting. This section explains the three pieces in turn: the radial basis used for diffraction, the time stepping scheme that combines them, and the population equations that must be solved alongside the field.

4.1 Quasi Discrete Hankel Transform for the Radial Coordinate

The beam has cylindrical symmetry, so the transverse Laplacian $\nabla_\perp^2$ acting on a field that depends only on r reduces to $\frac{1}{r}\frac{\partial}{\partial r}\big(r\frac{\partial}{\partial r}\big)$. The natural basis for this operator is not the Fourier basis but the Hankel transform of order zero, since a plane diffracting wave with cylindrical symmetry is a superposition of Bessel functions $J_0(\rho r)$ rather than complex exponentials. In this basis, diffraction becomes a simple multiplication by a phase factor, exactly as dispersion becomes a multiplication in the temporal Fourier domain.

The continuous Hankel transform cannot be evaluated on a computer, so the code uses the quasi discrete Hankel transform (QDHT) of Guizar-Sicairos and Gutierrez-Vega. The key idea is to sample the field not on a uniform radial grid, but at the zeros $j_1, j_2, \ldots, j_N$ of the Bessel function J_0. On this specific grid, the forward and inverse Hankel transforms both collapse to the same linear transformation, represented by a single symmetric matrix Y:

$$Y_{mn} = \frac{2}{j_N}\,\frac{J_0(j_m j_n/j_N)}{J_1(j_m)^2} \tag{46}$$

Multiplying a field sampled at the radii $r_m = j_m R/j_N$ by Y gives its Hankel transform sampled at the spatial frequencies $\rho_m = j_m/R$, and multiplying by Y a second time returns the original field, since Y is self-inverse. This is exactly what `build_grids` constructs in Algorithm 8, step 2, and what `LinearOperator.half_diffraction` uses in Algorithm 11, applying Y before and after multiplying by the free space diffraction phase $\exp\big(-i\rho^2 dz/(2k_0)\big)$.

Note for revision. The derivation of why sampling at the Bessel zeros makes the discrete transform exact, rather than merely approximate, follows from the orthogonality relation of J_0 over $[0, R]$ with these specific nodes. If you want the full derivation from the continuous Hankel transform included here for completeness, let me know and I will add it with the original Guizar-Sicairos and Gutierrez-Vega, Optics Letters 29, 1978 (2004) reference.

4.2 Strang Splitting of the Propagation Step

The full propagation equation of Section 11 can be written schematically as $\partial_z u = (\mathcal{L}+\mathcal{N})u$, where $\mathcal{L}$ groups the linear, z-independent operators, diffraction and dispersion, and $\mathcal{N}$ groups the nonlinear terms, Kerr, self-steepening, and the two loss channels. Solving $\mathcal{L}$ alone and $\mathcal{N}$ alone are both computationally cheap, diffraction and dispersion are diagonal in the Hankel and Fourier bases respectively, and the nonlinear term is diagonal in real space. Solving $\mathcal{L}+\mathcal{N}$ together is not.

Strang splitting exploits this by approximating the exact evolution operator $e^{(\mathcal{L}+\mathcal{N})dz}$ with the symmetric product:

$$e^{(\mathcal{L}+\mathcal{N})dz} \approx e^{\mathcal{L}\,dz/2}\, e^{\mathcal{N}\,dz}\, e^{\mathcal{L}\,dz/2} \tag{47}$$

Because the operators do not commute, this product differs from the exact evolution, but the error is only third order in dz per step, compared to second order for the naive non symmetric splitting $e^{\mathcal{L}dz}e^{\mathcal{N}dz}$. This is what `Integrator.step`, Algorithm 15, implements: a half diffraction step, a half dispersion step, the full nonlinear step, then dispersion and diffraction again in reverse order. The code further splits the linear half step itself into diffraction and dispersion applied one after the other, which introduces a smaller additional splitting error between these two commuting free space operators, negligible compared to the Kerr and dispersion length scales of the problem.

Inside the nonlinear half step, the code separates two things that behave very differently. The absorption coefficient $\alpha_{ion} + \sigma_\omega N/2$, from photoionization loss and plasma absorption, only reduces the amplitude of u without exchanging energy between frequency components, so it is applied exactly as a multiplicative exponential factor $u \to u\, e^{-\alpha\, dz/2}$ around the rest of the nonlinear step. The remaining part, Kerr, self-steepening, and the dispersive part of the plasma response, is a genuine differential equation in z and is integrated with a fourth order Runge-Kutta scheme over the nonlinear substep, following exactly the four stage update visible in `Integrator.step`.

4.3 Population Rate Equations on the GPU

The rate equations of Section 11.2 are solved separately from the field envelope, using an explicit time marching scheme along the temporal axis at fixed radius, before every propagation step. Because the equations for N and N_{STE} are stiff, the photoionization and avalanche rates can change by orders of magnitude within a single time sample, a naive explicit Euler step would require an impractically small Δt. The code avoids this by treating the coupled pair as a locally linear system over each time interval $[t_j, t_{j+1}]$,

$$\frac{dx}{dt} = S + Lx \tag{48}$$

with S and L evaluated from the trapezoidal average of the intensity dependent rates at t_j and t_{j+1}, exactly as detailed in Algorithm 7. This linear equation has the closed form solution:

$$x(t_j + \Delta t) = x(t_j)\, e^{L\Delta t} + S\, \Delta t\, \varphi_1(L\Delta t), \qquad \varphi_1(z) = \frac{e^z - 1}{z} \tag{49}$$

detailed in Algorithm 5, with a Taylor fallback for φ_1 when $L\Delta t$ is close to zero to avoid a $0/0$ division. This is an exact integrator for the frozen coefficient problem, it does not accumulate truncation error from the time stepping itself, only from the assumption that S and L are constant over each short interval, which is a much better approximation than freezing the full nonlinear right hand side of a generic ODE.

4.4 The Main Propagation Loop

Figure 9 puts the three preceding subsections together into the single loop that `Integrator.propagate` runs once per z step, combining Algorithms 14 and 15.

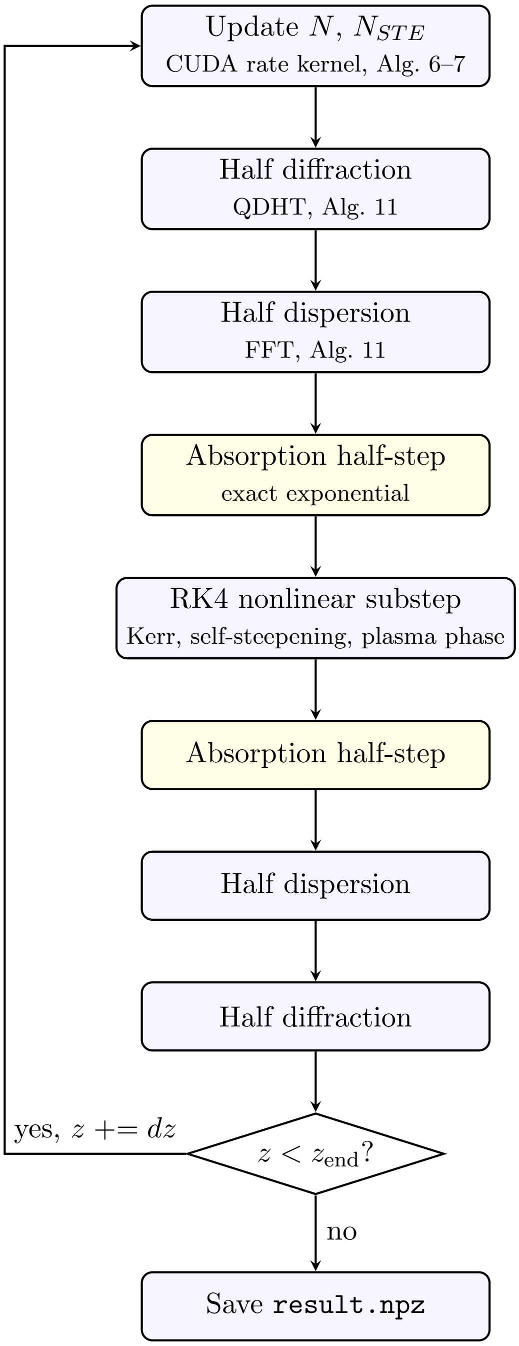


Figure 9: One iteration of the main propagation loop in `Integrator.propagate`. Every z step first advances the carrier populations along the full time axis at fixed r (Section 13), then applies the symmetric Strang sequence of diffraction, dispersion, and nonlinear substeps derived above to the field envelope. Diagnostics are recorded every `save_stride` steps (omitted here for clarity) before the loop repeats until z reaches the end of the propagation window.

5 GPU Acceleration and CUDA Interpolation

5.1 Why the Grid Needs a GPU

The propagation grid has of order 10^3 radial points, 4×10^3 time points, and 10^4 to 10^5 propagation steps, so the field envelope alone is an array of several million complex numbers updated at every step. Every step also requires two matrix multiplications with the QDHT matrix Y, two FFTs along the time axis, and a full evaluation of the nonlinear right hand side at four Runge-Kutta stages. On a CPU this would take from several hours to days for a single run. The code offloads every one of these operations to the GPU through `cupy`, a drop-in replacement for `numpy` that runs the same array operations on CUDA hardware, so the matrix products, FFTs, and elementwise nonlinear evaluations all execute in parallel across thousands of GPU cores instead of sequentially on the CPU.

5.2 The Photoionization Rate Table

The Keldysh photoionization rate W_{PI} derived in Section 3 involves an infinite sum of Dawson functions and complete elliptic integrals, described in Algorithm 3. This is far too expensive to evaluate at every grid point and every time step during propagation, it would dominate the runtime by orders of magnitude. Instead, the code evaluates W_{PI} once, before propagation starts, on a dense logarithmically spaced grid of intensities spanning the full range the simulation can reach, and fits a monotone cubic Hermite interpolant (PCHIP) through these points, as shown in Algorithm 9, step 3. The result is a lookup table of 4096 points in $\log_{10} I$, stored on the GPU, one table for the valence to conduction band gap U_i and a second table for the shallower self-trapped exciton gap U_s.

During propagation, whenever the code needs W_{PI} at some intensity, it no longer evaluates the analytical Keldysh formula, it looks up the table and interpolates linearly between the two nearest samples in log intensity space, described in Algorithm 4. Because the table is sampled logarithmically and the Keldysh rate varies smoothly on a logarithmic intensity scale, this reduces a computation that would otherwise involve dozens of transcendental function evaluations down to two table reads and one multiplication, at the cost of a small and controlled interpolation error.

5.3 The CUDA Kernel for the Rate Equations

The interpolation above is not only used from the array level, it is also compiled directly into a custom CUDA kernel written in the raw CUDA C string `_RATE_KERNEL_SRC` in `NewSim3juillet.py` and launched through `cupy.RawKernel`. This is a deliberate departure from the array based style used everywhere else in the code, and it exists because the rate equations of Section 11.2 must be integrated sequentially in time at every radius, one time sample depends on the previous one, so this step cannot be vectorized as a single array operation the way the FFTs and the QDHT product can.

The kernel assigns one CUDA thread to each radial point, following Algorithm 6. Each thread then walks the entire time axis in a tight loop, from t_1 to t_{N_t}, evaluating the interpolated rate `interpolate_pi_rate` at every step and advancing N and N_{STE} with the exact exponential update of Algorithm 7. Because thousands of radial points are independent of one another, thousands of these sequential time loops run concurrently on the GPU, one per thread, which is the only way to make an inherently sequential recurrence relation fast, parallelize across the dimension that has no sequential dependency, here the radius, instead of trying to parallelize the dimension that does, here the time.

Note for revision. The exact number of CUDA threads per block, currently `threads=256` with `blocks` computed to cover `Nr`, is a practical GPU occupancy choice rather than a physical one. If you want a short justification of this block size added here based on the GPU model you run this on, tell me which GPU you use and I will add the reasoning.

Update relative to the July solver. `deliverable (3)/sim/filament_sim.py` extends the kernel with the optional self-trapped exciton decay term of Section 11.2, adding an `inv_tau_ste` argument that subtracts N_{STE}/τ_{STE} from the exponential update of N_{STE}

when enabled, with no effect on N. The kernel itself could not be executed in the environment that produced `deliverable (3)`, since it has no GPU or `cupy` installed. Instead, `NonlinearOperator.split` was checked with a pure NumPy mock of `cupy` exercising all six ablation switches of Section 11.1, confirming that disabling every nonlinear term simultaneously gives an exactly zero nonlinear right hand side, that each switch changes the output as expected without producing non-finite values, and that disabling self-steepening forces the operator $\hat{T}$ to the identity. This is a unit level check of the Python and array logic, not a validation of the CUDA kernel itself, which still needs to be run once on GPU hardware before the results are trusted for production use.

5.4 Summary of the GPU Division of Labor

To summarize, three different accelerations coexist in the code, each solving a different bottleneck. The QDHT matrix product and the FFTs are handled by `cupy`'s built in linear algebra and FFT routines, which are already GPU native and require no custom code. The photoionization rate is precomputed once on the CPU with `scipy`, then transferred to the GPU as a lookup table, trading an expensive repeated computation for a cheap repeated memory read. The population rate equations, which are sequential in time and cannot be expressed as a single array operation, are handled by a hand written CUDA kernel that exploits parallelism across the radial coordinate instead. Together these three choices are what make it possible to run a full three dimensional filamentation simulation, two transverse plus time, over tens of thousands of propagation steps, in a runtime of minutes rather than days.

6 Interferometric Diagnostics and the Abel Transform

The propagation code produces a three dimensional map of the free electron density, the self-trapped exciton density, and the intensity, but none of these quantities can be measured directly inside a bulk sample. What can be measured is the phase shift accumulated by a probe beam that crosses the excited region, using a Nomarski interferometer. This section derives how that phase shift relates to the refractive index change predicted by the simulation, and describes how `unified_filament_slider_v3.py` implements both directions of this link, from a measured interferogram back to a refractive index profile, and from a simulated refractive index profile forward to a predicted interferogram. The physical setup and the mathematics of the Abel transform below are identical whether the excited medium is air or fused silica, only the propagation equation that produces the refractive index change differs, and that equation was already given in Section 11 for SiO_2.

6.1 Two-Plane-Wave Interferometry

The Nomarski interferometer can be understood as a configuration that allows retrieval of phase information about a medium under investigation. We derive the fundamental equations governing the interference pattern produced by two plane waves with different propagation directions.

6.1.1 Field Configuration and Wave Vectors

Consider two plane waves propagating in the same medium with refractive index η_i:

$$E_1 = \underline{E_0}\, e^{i(\vec{k}_1 \cdot \vec{r} - \omega t)} \tag{50}$$

$$E_2 = \underline{E_0}\, e^{i(\vec{k}_2 \cdot \vec{r} - \omega t)} \tag{51}$$

where the wave vectors are defined as:

$$\vec{k}_1 = \frac{2\pi}{\lambda}\eta_i \begin{pmatrix} \sin\theta \\ 0 \\ \cos\theta \end{pmatrix} = k \begin{pmatrix} \sin\theta \\ 0 \\ \cos\theta \end{pmatrix}, \qquad \vec{k}_2 = k \begin{pmatrix} -\sin\theta \\ 0 \\ \cos\theta \end{pmatrix} \tag{52}$$

These two plane waves propagate symmetrically with respect to the normal axis, making an angle 2θ with each other. To account for an arbitrary orientation of the fringes around the z axis, we introduce a rotation of angle ϕ, which turns the wave vectors into $k(\cos\phi\sin\theta, \sin\phi\sin\theta, \cos\theta)$ and its mirror image.

6.1.2 Intensity Distribution and Fringe Spacing

Setting the observation screen at $z = 0$, the intensity is:

$$I(x, y) = I_1 + I_2 + \sqrt{I_1 I_2}\cos\left((\vec{k}_1 - \vec{k}_2)\cdot\vec{r}\right) \tag{53}$$

For equal amplitudes and after computing the phase difference $(\vec{k}_1 - \vec{k}_2)\cdot\vec{r} = 2k\sin\theta(x\cos\phi + y\sin\phi)$, this becomes:

$$\boxed{I(x, y) = 2I_0(x, y)\left[1 + \cos\left(2\pi\nu_0(x\cos\phi + y\sin\phi)\right)\right]}, \qquad \nu_0 = \frac{2\sin\theta}{\lambda} \tag{54}$$

where ν_0 is the spatial-carrier frequency of the fringes, set purely by the crossing angle 2θ of the two interfering beams, and independent of anything happening inside the medium.

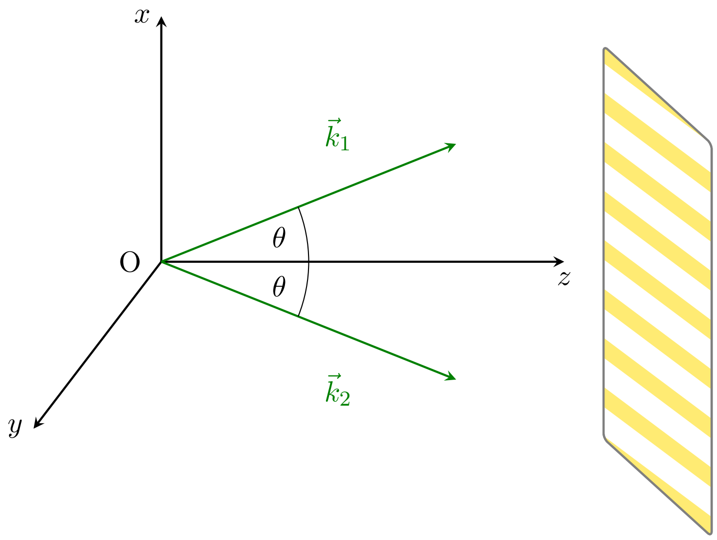


Figure 10: Interference of two plane waves propagating symmetrically with an angle 2θ, creating an interference pattern on the screen (shown for $\phi = 0$).

6.2 Phase and Amplitude Extraction via Fourier Analysis

Nomarski interferograms read $I(x, y) = a(x, y) + b(x, y)\cos\left[2\pi\nu_0 x + \varphi(x, y)\right]$, where $a(x, y)$ is the background term, $b(x, y)$ the amplitude modulation term, and $\varphi(x, y)$ the phase of interest, related to the line integrated electron density by $\varphi(x, y) \propto \int n_e(x, y, z)\,\mathrm{d}z$. In conventional contour-fringe interferometry, obtained when $\nu_0 = 0$, the intensity reads $I = a + b\cos\varphi$, in which a, b, and φ are mixed, which limits accuracy, and the parity of the cosine prevents determining the sign of φ.

The Takeda method [14] overcomes both limitations through spatial homodyne demodulation. As in classical heterodyne detection, the signal of interest is carried by an oscillating reference, here the spatial carrier ν_0 generated by the tilted incidence angle of the interferometer arms. Writing the cosine with Euler's formula gives $I(x,y) = a(x,y) + c(x,y)e^{i2\pi\nu_0 x} + c^*(x,y)e^{-i2\pi\nu_0 x}$ with $c(x,y) = \frac{1}{2}b(x,y)e^{i\varphi(x,y)}$. Bandpass filtering around $+\nu_0$ in the Fourier plane isolates the sideband carrying $c(x,y)$ alone, from which b and φ are recovered respectively as the modulus and the argument.

This linear separation also improves sensitivity to weak phase variations. Contour interferometry responds quadratically to a small phase, $\Delta I_{contour} \approx \frac{b}{2}\varphi^2$, because small variations sit near the extremum of a cosine, whereas the Takeda method responds linearly, $\Delta I_{Takeda} \approx -b\sin(2\pi\nu_0 x)\,\varphi$, which is what gives it sub-wavelength phase sensitivity [14]. Applied to a reference interferogram (no plasma or no excitation) and a measurement interferogram, the ratio of the two filtered and inverse transformed signals cancels the carrier and any residual optical misalignment:

$$z(x,y) \equiv \frac{I_{\text{meas,filt}}(x,y)}{I_{\text{ref,filt}}(x,y)} = \frac{b_2(x,y)}{b_1(x,y)}\, e^{i\varphi(x,y)}, \qquad \boxed{\varphi(x,y) = \text{atan2}\left(\text{Im}\{(z)\}, \text{Re}\{(z)\}\right)} \tag{55}$$

6.3 Resolution Limits

Two independent limits set the finest feature that can be resolved. The Nyquist limit comes from the pixelated sensor. The shortest oscillation observable without aliasing must span at least two pixels, giving $\nu_N = 1/(2\Delta x)$ with Δx the pixel size. The optical limit comes from diffraction at the sample itself. A feature of size d diffracts light at an angle $\sin\theta = \lambda/d$, as illustrated in Figure 12, and this diffracted light is only collected by the imaging lens if $\sin\theta \le NA$.

6.3.1 Contrast at the Nyquist Limit

The Nyquist frequency $\nu_N = 1/(2\Delta x)$ only says that a fringe period of exactly two pixels can be represented without aliasing. It says nothing about how much contrast survives once the continuous fringe pattern is integrated over the finite area of each pixel, which is what a real sensor does. For fringes tilted at 45°, the intensity recorded by the pixel at (n_x, n_y) is the average of the continuous pattern $\cos^2(2\pi f_x x + 2\pi f_y y + \phi)$ over that pixel:

$$\begin{aligned} I(n_x,n_y) &= \int_{n_x-\frac{1}{2}}^{n_x+\frac{1}{2}} \int_{n_y-\frac{1}{2}}^{n_y+\frac{1}{2}} \cos^2(2\pi f_x x + 2\pi f_y y + \phi)\, dx\, dy \\ &= \frac{1}{2} + \frac{1}{2}\cos(2\pi f_x n_x + 2\pi f_y n_y + \phi) \left[\frac{\sin(\pi f_x)}{\pi f_x}\right] \left[\frac{\sin(\pi f_y)}{\pi f_y}\right] \end{aligned} \tag{56}$$

At the Nyquist limit itself, $f_x = f_y = \frac{1}{2}$, this becomes:

$$I(n_x,n_y) = \frac{1}{2} + \frac{2}{\pi^2}\cos\left(\pi(n_x+n_y) + \phi\right) \tag{57}$$

so successive pixels alternate between $\frac{1}{2} + \frac{2}{\pi^2}$ and $\frac{1}{2} - \frac{2}{\pi^2}$, a modulation of only about ± 0.20 around the mean instead of the full ± 1 of the underlying fringe. Sampling exactly at ν_N is therefore technically alias free but already badly starved of contrast, which is why in practice the sensor should be operated with some margin below ν_N rather than pushed all the way to it.

6.3.2 Separation of Interference Orders in Fourier Space

Filtering a single sideband, as in the Takeda method above, only works if that sideband does not overlap the other two lobes present in the Fourier transform of a recorded interferogram, the zero order at the origin and the conjugate sideband on the other side. The object beam itself has a finite bandwidth set by the optical resolution: for a sensor image magnified by M_{tot}, the smallest resolvable feature is $R_s = M_{tot}\lambda/(2NA)$ by the Abbe criterion, so the object spectrum $\hat{c}(\nu_x, \nu_y)$ is confined to a disk of radius

$$r_f = \frac{1}{2R_s} = \frac{NA}{M_{tot}\lambda}. \tag{58}$$

The zero order term is proportional to $|c|^2$, whose Fourier transform is the autocorrelation $\hat{c} \circledast \hat{c}^*$, so it occupies the larger disk of radius $2r_f$ centered at the origin. The two sidebands sit at $\pm\nu_0$, with ν_0 the carrier frequency of Eq. (54), each carrying a copy of the object spectrum of radius r_f. Center-to-center, non-overlap of the sideband at ν_0 with the central autocorrelation blob requires the separation to exceed the sum of the two radii, giving the same order of magnitude condition as the general one-dimensional argument for off-axis holography, $\nu_0 \gtrsim 3r_f$. Kimura's more detailed two-dimensional treatment of the 45°-tilted configuration used in this setup refines this into two explicit design conditions,

$$\sqrt{3}\, r_f < \nu_0, \qquad \nu_0 + \sqrt{2}\, r_f < \sqrt{2}\, \nu_N, \tag{59}$$

the first ensuring the sideband and the zero order stay separated, the second ensuring the sideband, once separated, still fits entirely inside the region the sensor can sample without aliasing. If the carrier frequency ν_0 is set too low for a given optical resolution r_f, the sideband bleeds into the zero order and no filter can cleanly separate them again, which caps how much optical resolution a given fringe spacing can support, and conversely how tight the fringe spacing must be pushed to preserve a given resolution.

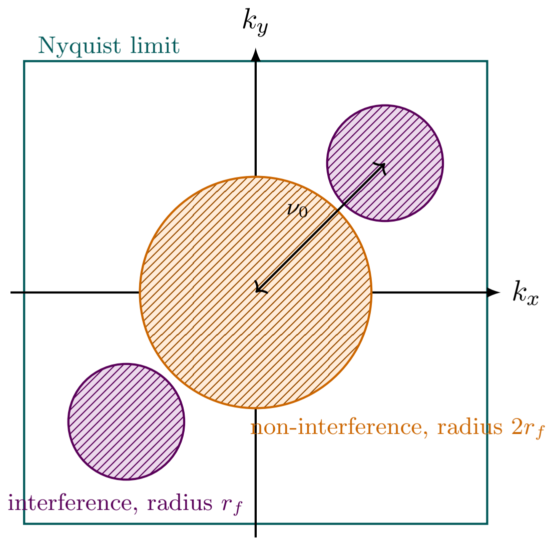


Figure 11: Layout of the interference orders in the Fourier plane of a recorded interferogram. The non-interference (zero order) blob of radius $2r_f$ sits at the origin, while the two interference sidebands of radius r_f sit at $\pm\nu_0$ along the carrier direction. Clean separation for filtering requires both sideband disks to stay outside the zero order disk and inside the square region the sensor can sample without aliasing, set by the Nyquist frequency ν_N.

Note for revision. The reference for this subsection is a chapter from what appears to be Kimura's thesis (file `partie_theorique_Kimura.pdf`), which I do not have full bibliographic details for beyond the filename. If you can give me the author's full name, the institution, and the year, I will complete the citation properly instead of leaving it as `kimura_thesis`.

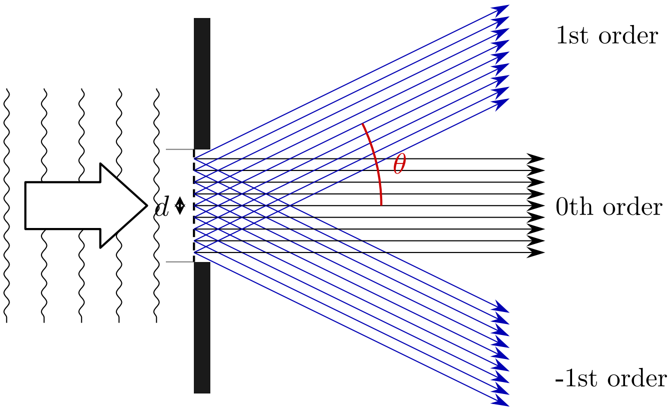


Figure 12: A feature of size d diffracts the probe beam into orders separated by $\sin\theta = \lambda/d$. Only orders within the lens acceptance angle are collected.

Inverting the problem gives the smallest resolvable feature for a lens of numerical aperture $NA = \sin\alpha$ in a coherent on axis configuration, $R_0 = \lambda/NA$, shown in Figure 13. After magnification by the imaging system, the highest spatial frequency actually delivered to the sensor is $\nu_{max} = NA/(M_{tot}\lambda)$, and the digital sampling must satisfy $\nu_N \geq \nu_{max}$ for the optical resolution to be preserved rather than lost to undersampling.

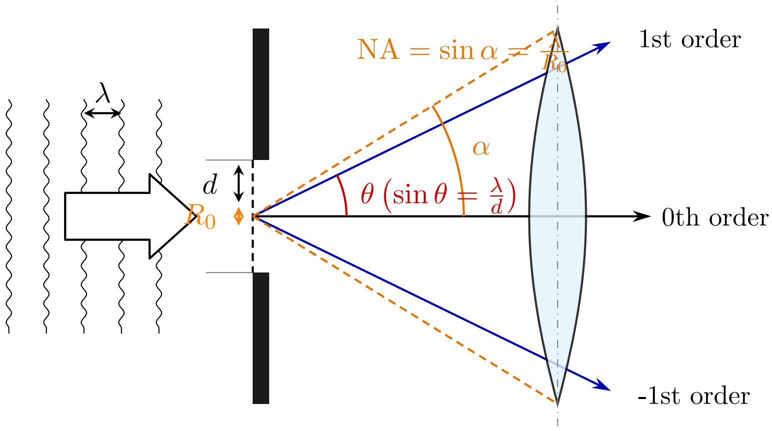


Figure 13: Smallest resolvable feature $R_0 = \lambda/NA$ for a lens of numerical aperture $NA = \sin\alpha$, obtained by requiring the first diffraction order to fall within the acceptance cone.

6.3.3 Sub-Pixel Correction of the Carrier Frequency

The Fourier transform method above assumes the carrier peak used to recenter the sideband, (u_0, v_0), is known exactly. In practice it is only known to the frequency resolution of the discrete Fourier transform. For a hologram sampled on an $N \times N$ grid, that resolution is $\Delta f_x = \Delta f_y = 1/N$, and nothing guarantees that the true carrier frequency falls exactly on a grid point [17]. If the true peak is offset from the nearest FFT bin by a fractional amount $(\Delta x, \Delta y)$, with $|\Delta x|, |\Delta y| < 0.5$ pixel in frequency space, shifting the sideband to that nearest bin instead of the true peak leaves a residual linear phase ramp across the whole reconstructed phase map:

$$\Delta\varphi(x, y) = \frac{2\pi}{N}\left(\Delta x\, x + \Delta y\, y\right). \tag{60}$$

This shows up as a uniform tilt superimposed on the recovered phase, easy to mistake for a real gradient in the plasma or the sample if left uncorrected.

The fix is to refine the peak location to sub-pixel accuracy before using it, by evaluating a local, up-sampled discrete Fourier transform in a small neighborhood around the coarse FFT maximum rather than trusting the coarse grid point itself. Writing the hologram as $H(X, Y)$ on the spatial grid $X = Y = [-N/2, \ldots, N/2-1]^T$, this local transform is the matrix product

$$F(U, V) = \exp\left(-i\tfrac{2\pi}{N} U X^T\right) H(X, Y)\, \exp\left(-i\tfrac{2\pi}{N} Y V^T\right), \tag{61}$$

evaluated only on a small set of frequencies U, V centered on the coarse peak (u_0, v_0) and spaced by $1/\alpha$ instead of 1, where $\alpha \sim 10$ is an up-sampling factor [17]. Because U and V only span a window of a couple of pixels around (u_0, v_0), this costs little compared with the full $N \times N$ transform even though it resolves the peak far more finely. The maximum of $|F(U, V)|$ in this window gives the refined peak $(u_0+\Delta x, v_0+\Delta y)$, from which $\Delta\varphi(x, y)$ above is computed and subtracted from the phase map obtained by the ordinary four step Fourier transform method, removing the spurious tilt without needing a higher resolution FFT of the entire hologram.

Note for revision. I have not seen this sub-pixel correction implemented anywhere in `unified_filament_slider_v3.py` or `abel_phase_explorer.py`. Both currently locate the carrier peak with a single coarse FFT maximum (`find_1storder_peak`) and rely on `fit_plane_from_mask` to remove any resulting tilt from a background region after the fact, which corrects the same symptom but only where a flat background region is actually available in the image. If sub-pixel peak refinement is not already planned, it may be worth adding directly at the peak-finding step rather than only patching the tilt afterward.

6.4 The Abel Transform

Consider a probe beam crossing the excited cylindrically symmetric region along y, at fixed x and z. The accumulated phase relative to a beam that has not crossed the excited region is:

$$\phi(x, z) = 2\left(\frac{2\pi}{\lambda_L}\right)\int_0^\infty [\eta(r, z) - 1]\, dy, \qquad r = \sqrt{x^2 + y^2} \tag{62}$$

Using the substitution $y = \sqrt{r^2 - x^2}$, this line integral becomes the forward Abel transform of the refractive index deviation $f(r) \equiv \eta(r, z) - 1$:

$$\boxed{F(x) = 2\int_x^\infty \frac{f(r)\, r}{\sqrt{r^2 - x^2}}\, dr}, \qquad F(x) \equiv \frac{\lambda_L}{4\pi}\phi(x, z) \tag{63}$$

This is consistent with equation (4) of reference [16]. The Abel transform can be inverted analytically. Applying the operator $\int_\rho^\infty x\, dx/\sqrt{x^2 - \rho^2}$ to both sides and using the identity $\int_\rho^r x\, dx/\sqrt{(r^2 - x^2)(x^2 - \rho^2)} = \pi/2$ gives, after differentiating with respect to ρ:

$$\boxed{f(\rho) = -\frac{1}{\pi}\int_\rho^\infty \frac{dF(x)/dx}{\sqrt{x^2 - \rho^2}}\, dx} \quad \Longrightarrow \quad \boxed{\eta(r, z) = 1 - \frac{1}{\pi}\left(\frac{\lambda_L}{2\pi}\right)\int_r^\infty \frac{d\phi(x, z)/dx}{\sqrt{x^2 - r^2}}\, dx} \tag{64}$$

The Abel transform is also a special case of the more general Radon transform, which describes the line integral of a function along an arbitrary ray at angle θ and impact parameter

ρ. For a rotationally symmetric function $f(r)$, the projection is the same at every angle, and choosing $\theta = 0$ turns the Radon integral into $\mathcal{R}[f](\rho) = 2\int_\rho^\infty f(r)\, r\, dr/\sqrt{r^2-\rho^2}$, exactly the forward Abel transform above. This equivalence is what justifies treating the interferometric phase, a line integrated quantity, as a projection that can be inverted whenever cylindrical symmetry holds.

6.5 Numerical Inversion by the Trapezoidal Rule

To invert the Abel transform numerically, the integral is approximated by summing trapezoidal areas over the integrand g,

$$A_k \approx \tfrac{1}{2}(x_{k+1}-x_k)\big[g(x_{k+1})+g(x_k)\big], \qquad g(x_k, r_i) = \frac{\big[dF(x)/dx\big]_{x=x_k}}{\sqrt{x_k^2 - r_i^2}}. \tag{65}$$

The derivative is itself approximated by a forward difference, $dF/dx|_{x_k} \approx [F(x_{k+1}) - F(x_k)]/\Delta x$, and the integration starts one pixel away from $x = r_i$ to avoid the mathematical singularity there. Substituting the forward difference into the trapezoidal sum, the step size Δx cancels exactly, leaving:

$$\boxed{f(r_i) = -\frac{1}{2\pi}\sum_{k=i+1}^{N-2}\left[\frac{F(x_{k+2})-F(x_{k+1})}{\sqrt{x_{k+1}^2 - r_i^2}} + \frac{F(x_{k+1})-F(x_k)}{\sqrt{x_k^2 - r_i^2}}\right]} \tag{66}$$

directly implementable without ever storing Δx as a separate factor.

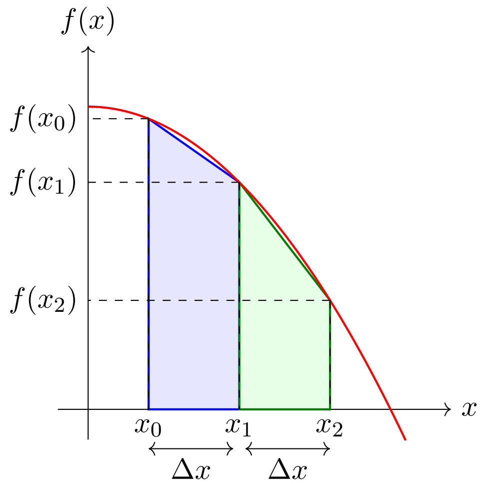


Figure 14: Illustration of the trapezoidal rule for numerical integration. The integral under the curve is approximated as a sum of trapezoidal areas.

6.6 Connecting the Simulation to the Diagnostic

The script `unified_filament_slider_v3.py` implements both directions of the link between the simulation and a real interferometric measurement, so that the two can be compared directly.

On the experimental side, `find_1storder_peak` locates the side band of a raw interferogram in Fourier space by masking out the central DC region and searching for the strongest remaining peak, implementing the Takeda demodulation step described above. `reconstruct` then crops a disk of radius set by the numerical aperture around that peak and inverse transforms it back to real space, giving the complex field $c(x, y)$ from which the phase is extracted. `fit_plane_from_mask`, applied over user selected background regions through `baseline_rects_to_mask`, removes any residual linear phase tilt from small optical misalignments that the reference division does not fully cancel, exactly the correction motivated at the end of the phase retrieval derivation above. `preprocess_side_full` chains these steps into the full pipeline from a raw pair of interferograms, signal and reference, to a clean two dimensional phase map.

On the simulation side, `build_abel_matrix` discretizes the forward Abel transform directly as a matrix acting on the non uniform radial grid produced by the QDHT, using the exact shell integral of $r/\sqrt{r^2 - x^2}$ over each radial bin rather than a naive trapezoidal sum on that grid, so that the same physical relation $F(x) = 2\int_x^\infty f(r)r\,dr/\sqrt{r^2 - x^2}$ used above for the inversion is applied here in the forward direction. `abel_phase_2d` assembles the total refractive index change Δn from every channel the propagation code tracks, the Drude plasma contribution from the free electron density, the Lorentz contribution from the self-trapped exciton density, the Kerr contribution from the local intensity, and a thermal contribution computed by `precompute_dn_thermal` from the energy deposited by trapped carriers relaxing into phonons, before applying the Abel matrix to predict the phase map a real interferometer would record for that simulated filament at a given experimental probe delay.

Note for revision. The thermal channel in `precompute_dn_thermal` assumes isochoric heating with a single phonon relaxation time `tau_ph_fs` and a fixed deposited energy per trapped carrier. I have not verified this against a reference for the thermo-optic coefficient dn/dT or the heat capacity values used. If you can point me to the source for these, I will add the derivation and the citation here.

References

[1] R. L. Fork, B. I. Greene, and C. V. Shank, *Generation of optical pulses shorter than 0.1 psec by colliding pulse mode locking*, Appl. Phys. Lett. (1981).

[2] A.H. Zewail, *Femtochemistry: Ultrafast Dynamics of the Chemical Bond*, World Scientific, Singapore (1994).

[3] J. Shah, *Ultrafast Spectroscopy of Semiconductors and Semiconductor Nanostructures*, Springer, Berlin Heidelberg New York (1996).

[4] S.S. Mao et al., *Dynamics of femtosecond laser interactions with dielectrics*, Appl. Phys. A 79, 1695–1709 (2004).

[5] N.M. Bulgakova, R. Stoian, A. Rosenfeld, *Laser-induced modification of transparent crystals and glasses*, Quantum Electron. 40 966 (2010).

[6] A. Couairon, E. Mével, L. Berge, G. Pénano, M. Quick, A. Guérin, and P. Morel, *Femtosecond filamentation in fused silica*, Phys. Rev. B 71, 125435 (2005).

[7] A. Couairon, A. Mysyrowicz, *Femtosecond filamentation in transparent media*, Phys. Rep. 441, 47–189 (2007).

[8] A.L. Gaeta, *Catastrophic collapse of ultrashort pulses*, Phys. Rev. Lett. 84, 3582 (2000).

[9] V.P. Kandidov, S.A. Shlenov, O.G. Kosareva, *Filamentation of high-power femtosecond laser radiation*, Quantum Electron. 39, 205–228 (2009).

[10] J.H. Marburger, *Self-focusing: Theory*, Prog. Quantum Electron. 4, 35–110 (1975).

[11] E.L. Dawes, J.H. Marburger, *Computer studies in self-focusing*, Phys. Rev. 179, 862–868 (1969).

[12] J.E. Rothenberg, *Pulse splitting during self-focusing in normally dispersive media*, Opt. Lett. 17, 583–585 (1992).

[13] D. Milam, *Review and assessment of measured values of the nonlinear refractive-index coefficient of fused silica*, Appl. Opt. 37, 546–550 (1998).

[14] M. Takeda, H. Ina, S. Kobayashi, *Fourier-transform method of fringe-pattern analysis for computer-based topography and interferometry*, J. Opt. Soc. Am. 72, 156–160 (1982).

[15] E.N. Leith, J. Upatnieks, *Reconstructed wavefronts and communication theory*, J. Opt. Soc. Am. 52, 1123–1130 (1962).

[16] B. Gonzalez-Izquierdo, H. Sakurai, K. Konishi, J. Yumoto, *Plasma electron density and plasma temperature characterization in air and solid targets: interferometry and shadowgraphy*, J. Appl. Phys. (2012).

[17] K. Khare, M. Butola, S. Rajora, *Fourier Optics and Computational Imaging*, Chapter 14: Digital Holography, Springer Nature Switzerland AG (2023).

[18] Kimura, *[thesis title and institution to be completed]*, theoretical background chapter, provided as `partie_theorique_Kimura.pdf`.

A Algorithm Listings

The pseudocode below gives the exact, line by line correspondence with `NewSim3juillet.py` referenced throughout Sections 11 to 13.

Subroutine SELLMEIER *(Fused-Silica Linear Refractive Index)*:

Input: Vacuum wavelength λ [m]

Output: Linear refractive index $n(\lambda)$

$\lambda_{\mu m} \leftarrow \lambda \times 10^6$ `// Convert to micrometres for tabulated coefficients`

$n^2 - 1 \leftarrow \sum_{i=1}^{3} \dfrac{B_i\,\lambda_{\mu m}^2}{\lambda_{\mu m}^2 - L_i^2}$ `// Three-term fused-silica Sellmeier sum`

return $n \leftarrow \sqrt{1 + (n^2 - 1)}$ `//` $\{B_i\}, \{L_i\}$`: fused-silica constants`

Algorithm 1: Linear index of fused silica (`n_sellmeier`).

Class CONFIG *(Simulation Parameters & Derived Physical Constants)*:

Data: Default attributes grouped by role: numerical grid ($\texttt{nz}, \texttt{Nt}, \texttt{N}, \ldots$); laser ($\lambda$, energy_uJ, w0, delta_t); material SiO_2 ($\texttt{n2}, \texttt{Ui\_eV}, \texttt{tau\_c}, \ldots$); STE channel (Us_eV); physics switches; probe (lambda_probe); adaptive-step controls

Method POSTINIT *(__post_init__): derive constants & grid quantities*:

Output: Instance augmented with derived physical constants and grid scalars

1. Angular Frequencies & Probe Optics:

$\omega_0 \leftarrow 2\pi c/\texttt{wavelength}$ // Pump carrier angular frequency

$\omega_{pr} \leftarrow 2\pi c/\texttt{lambda\_probe}$ // Probe angular frequency

$n_{0,pr} \leftarrow \textsc{Sellmeier}(\texttt{lambda\_probe})$ // Probe linear index

$\rho_{c,pr} \leftarrow (\varepsilon_0 m_e \omega_{pr}^2/e^2) \times 10^{-6}$ // Probe critical plasma density [cm^{-3}]

2. Pump Linear Optics (Sellmeier):

$n_0 \leftarrow \textsc{Sellmeier}(\texttt{wavelength})$ // Pump linear index

$k_0 \leftarrow (2\pi/\texttt{wavelength})\, n_0$ // Pump wavenumber

3. Energy Gaps & Effective Masses:

$U_i \leftarrow \texttt{Ui\_eV} \cdot e$ // Valence bandgap [J]

$U_s \leftarrow \texttt{Us\_eV} \cdot e$ // STE gap [J]

$m_{eff} \leftarrow \texttt{meff\_rel} \cdot m_e$ // Reduced effective mass (Keldysh)

$m_D \leftarrow \texttt{meff\_drude\_rel} \cdot m_e$ // Drude effective mass

4. Kerr & Raman Constants:

$\chi^{(3)} \leftarrow \frac{4}{3}\varepsilon_0 n_0^2 c\, \texttt{n2}$ // Third-order susceptibility from n_2

$\Omega_R^2 \leftarrow 1/\texttt{tau\_s}^2 + 1/\texttt{tau\_d}^2$ // Raman oscillator resonance frequency^2

5. Longitudinal Grid:

$\Delta z \leftarrow (\texttt{end} - \texttt{begin})/\texttt{nz}$ // Uniform propagation step

$b \leftarrow k_0\, \texttt{w0}^2$ // Confocal parameter (twice the Rayleigh range)

if *dz_init is unset* **then**

  $\texttt{dz\_init} \leftarrow \min(\Delta z, \texttt{dz\_max})$ // Initial adaptive-step ceiling

end

6. Peak Intensity & Field Amplitude:

if *peak_power_W is provided* **then**

  $I_0 \leftarrow 2\,\texttt{peak\_power\_W}/(\pi\, \texttt{w0}^2)$ // Gaussian peak intensity from peak power

else

  $I_0 \leftarrow 2\,\mathcal{E}_p/(\pi\, \texttt{w0}^2\, \texttt{delta\_t})$ // Pulse energy $\mathcal{E}_p = \texttt{energy\_uJ} \cdot 10^{-6}$

end

$I_0^{\mathrm{cm}} \leftarrow I_0 \times 10^{-4}$ // W/m^2 $\rightarrow$ W/cm^2 (Keldysh LUT input)

$E_0 \leftarrow \sqrt{2 I_0/(n_0 c\, \varepsilon_0)}$ // Peak electric-field amplitude

7. Radial Sampling Diagnostic:

$\Delta r \leftarrow \texttt{R\_factor}\, \texttt{w0}/(\texttt{N} - 1)$ // Radial grid spacing

$p_{w_0} \leftarrow \texttt{w0}/\Delta r$ // Radial points per beam waist w_0

if $p_{w_0} < 3$ **then**

  warn "radial under-sampling" // QDHT accuracy compromised; raise N

end

else if $p_{w_0} < 6$ **then**

  warn "marginal radial sampling" // Borderline transverse resolution

end

Algorithm 2: Simulation configuration and derived constants (Config).

Class KeldyshSiO2 *(Solid-State Keldysh Photoionization Model)*:

Method Constructor *(__init__): store parameters & derive constants*:

Input: Wavelength λ; gap `Ui_eV`; effective mass m_{eff}; linear index n_0; series length `N_sum`

Output: Instance holding $\{U, m_{eff}, n_0, \omega, \hbar, \texttt{N\_sum}\}$

$U \leftarrow \texttt{Ui\_eV} \cdot e$ `// Bandgap energy [J]`

$\omega \leftarrow 2\pi c/\lambda$ `// Carrier angular frequency`

Store $m_{eff}, n_0, \hbar, \texttt{N\_sum}$ as instance attributes `// Direct parameter retention (`$\hbar$`: SI constant)`

Method Rate *(rate): analytical photoionization rate*:

Input: Intensity I [W/cm^2]

Output: Photoionization rate $W_{PI}(I)$ [cm^{-3} s^{-1}]

1. Field & Keldysh Parameter:

$I_{SI} \leftarrow I \times 10^4$ `// W/cm`$^2 \rightarrow$ `W/m`^{2}

$E \leftarrow \sqrt{2I_{SI}/(n_0 c\,\varepsilon_0)}$ `// Peak field driving the ionization`

$\gamma \leftarrow \max\big(\omega\sqrt{m_{eff}U}/(eE),\ 10^{-12}\big)$ `// Keldysh parameter; floored to avoid` $E \to 0$ `singularity`

$\gamma_1 \leftarrow \gamma/\sqrt{1+\gamma^2}$ `// Tunneling-side modulus`

$\gamma_2 \leftarrow 1/\sqrt{1+\gamma^2}$ `// Multiphoton-side modulus (`$\gamma_1^2 + \gamma_2^2 = 1$`)`

2. Complete Elliptic Integrals (Dual Modulus):

$K_1, E_1 \leftarrow \text{EllipticK}(\gamma_1^2),\ \text{EllipticE}(\gamma_1^2)$ `// Parameter` $m = \gamma_1^2$

$K_2, E_2 \leftarrow \text{EllipticK}(\gamma_2^2),\ \text{EllipticE}(\gamma_2^2)$ `// Parameter` $m = \gamma_2^2$

3. Effective Ionization Order:

$x \leftarrow 2UE_2/(\pi\gamma_1\hbar\omega)$ `// Effective bandgap in dressed-photon units (MPI order)`

4. Dawson Series (Keldysh Q-function):

for $n \leftarrow 0$ **to** *N_sum* $- 1$ **do**

  $a_n \leftarrow \pi\sqrt{\big(2\lfloor x+1\rfloor - 2x + n\big)/(2K_2E_2)}$ `// Argument of the` n`-th quasi-energy channel`

  $s_n \leftarrow \exp\big(-\pi n(K_1 - E_1)/E_2\big) \cdot \text{Dawson}(a_n)$ `// Channel weight` $\times$ `Dawson integral`

end

$\Sigma \leftarrow \sum_{n=0}^{\texttt{N\_sum}-1} s_n$ `// Sum over ionization channels`

$Q \leftarrow \sqrt{\pi/(2K_2)}\,\Sigma$ `// Keldysh` Q`-function`

5. Rate Assembly:

$\text{pref} \leftarrow \dfrac{2\omega}{9\pi}\left(\dfrac{\omega m_{eff}}{\hbar\gamma_1}\right)^{3/2}$ `// Keldysh prefactor [m`$^{-3}$ `s`$^{-1}$`]`

$W_{PI} \leftarrow \text{pref} \cdot Q \cdot \exp\big(-\pi\lfloor x+1\rfloor(K_1 - E_1)/E_2\big) \times 10^{-6}$ `// Exponential MPI/tunnel factor;` $\times 10^{-6}$`:` `m`$^{-3} \rightarrow$ `cm`$^{-3}$

return W_{PI}

Algorithm 3: Solid-state Keldysh photoionization model (`KeldyshSiO2`).

Function InterpolatePIRate *(Log-Linear LUT Lookup of W_{PI})*:

Input: Intensity I; rate table T of size N; log-grid origin $\log I_{min}$; inverse log-step $\Delta_{\log}^{-1}$; bounds I_{min}, I_{max}

Output: Interpolated photoionization rate $W_{PI}(I)$

1. Sanitization & Clamping:

if $\neg\, isfinite(I) \vee I < 0$ **then**

  $I \leftarrow 0$ `// Reject NaN/Inf/negative intensity`

end

$I \leftarrow \mathrm{Clip}(I, I_{min}, I_{max})$ `// Restrict to tabulated range`

2. Fractional Table Index:

$f_{idx} \leftarrow (\log_{10} I - \log I_{min})\, \Delta_{\log}^{-1}$ `// Map onto uniform log-intensity grid`

$idx \leftarrow \mathrm{Clip}(\lfloor f_{idx} \rfloor,\ 0,\ N-2)$ `// Keep` $idx{+}1$ `valid for interpolation`

3. Log-Linear Interpolation:

$W_{PI} \leftarrow T[idx] + (f_{idx} - idx)\,(T[idx{+}1] - T[idx])$ `// Linear blend between adjacent samples`

4. Output Sanitization:

if $\neg\, isfinite(W_{PI}) \vee W_{PI} < 0$ **then**

  return 0 `// Guard against invalid rate`

end

return W_{PI}

3. Output Sanitization:

if $\neg\, isfinite(x_{new}) \vee x_{new} < 0$ **then**

  return 0 `// Enforce physical non-negativity`

end

return x_{new}

Algorithm 4: Exact exponential integrator for $dx/dt = S + Lx$, i.e. $x(\Delta t) = x_0 e^{L\Delta t} + S\Delta t\, \varphi_1(L\Delta t)$ with $\varphi_1(z) = (e^z - 1)/z$, Taylor fallback for small $|z|$ (`exact_exp_step`).

Function ExactExpStep *(Exact Exponential Integrator for $\dot{x} = S + Lx$)*:

Input: Population x; source S; linear coefficient L; time step Δt

Output: Updated population x_{new}

1. Exponential φ_1 Function:

$L_\Delta \leftarrow L\,\Delta t$ `// Dimensionless linear exponent`

if $|L_\Delta| > 10^{-6}$ **then**

  $\varphi_1 \leftarrow (e^{L_\Delta} - 1)/L_\Delta$ `// Exact` $\varphi_1(z) = (e^z - 1)/z$

else

  $\varphi_1 \leftarrow 1 + L_\Delta\big(\frac{1}{2} + L_\Delta(\frac{1}{6} + L_\Delta/24)\big)$ `// Taylor fallback, avoids` $0/0$ `at small` $|L_\Delta|$

end

2. Exact Update:

$x_{new} \leftarrow e^{L_\Delta}\,x + S\,\Delta t\,\varphi_1$ `// Closed-form solution over` Δt

if $\neg\,\mathit{isfinite}(x_{new}) \vee x_{new} < 0$ **then**

  return 0 `// Enforce physical non-negativity`

end

return x_{new}

Algorithm 5: Exact exponential integrator for $dx/dt = S + Lx$, i.e. $x(\Delta t) = x_0 e^{L\Delta t} + S\Delta t\,\varphi_1(L\Delta t)$ with $\varphi_1(z) = (e^z - 1)/z$ (Taylor for small $|z|$).

Function SolveRateEquationKernel *(Part 1/2: CUDA Setup & $t = 0$ Initialization)*:

Input: Field envelope $\mathcal{E}$; density arrays n_e, n_s; grid sizes N_r, N_t; step Δt; field-to-intensity factor κ; avalanche coefficients β_g, β_s; neutral density n_a; recombination rate τ_r^{-1}; STE flag E_{ste}; PI tables T_g (CB), T_s (STE) with shared log-grid parameters

Output: Per-thread registers initialized at $t = 0$; time-stepping loop launched

1. Thread Mapping (one thread $\rightarrow$ one radial point):

```
r_i ← blockDim.x · blockIdx.x + threadIdx.x            // Global radial index
if r_i ≥ N_r then
    return                       // Early exit for out-of-bound threads
end
base ← r_i · N_t                 // Row offset into flattened (r, t) arrays
```

2. Initial Densities at $t = 0$:

```
n_e ← n_e[base],   n_s ← n_s[base]          // Load into local registers
if ¬ isfinite(n_e) ∨ n_e < 0 then
    n_e ← 0
end
if ¬ isfinite(n_s) ∨ n_s < 0 then
    n_s ← 0                                 // Sanitize seed densities
end
```

3. Initial Intensity & PI Rates:

```
f_0 ← E[base]                        // Complex field envelope at t = 0
I_c ← (f_{0,x}^2 + f_{0,y}^2) κ                  // Intensity = |E|^2 κ
W_c ← InterpolatePIRate(I_c, T_g, ...)       // CB photoionization rate
if E_ste ≠ 0 then
    W_c^s ← InterpolatePIRate(I_c, T_s, ...)  // STE photoionization rate
else
    W_c^s ← 0                                 // STE channel disabled
end
```

4. Time-Stepping Loop:

```
for j ← 0 to N_t − 2 do
    Execute TimeStepCore (Algorithm 7)       // Sequential march over
     temporal grid
end
```

5. Next-Step Field, Intensity & PI Rates:

```
f_1 ← E[base + j + 1]                          // Envelope at step j+1
I_n ← (f_{1,x}^2 + f_{1,y}^2) κ                   // Next-step intensity
W_n ← InterpolatePIRate(I_n, T_g, ...)       // CB photoionization rate
if E_ste ≠ 0 then
    W_n^s ← InterpolatePIRate(I_n, T_s, ...)  // STE photoionization rate
else
    W_n^s ← 0                                 // STE channel disabled
end
```

Algorithm 6: Two-population plasma rate-equation solver (Part 1/2): thread mapping, validation, $t = 0$ initialization (`solve_rate_equation_kernel`).

Function TIMESTEPCORE *(Part 2/2: Trapezoidal Step $j \to j{+}1$)*:

Input: Thread registers $base, n_e, n_s, I_c, W_c, W_c^s$; step index j

Output: Global memory at $base + j + 1$ written; registers advanced

1. Trapezoidal Averaging over $[j, j{+}1]$:

$I_{avg} \leftarrow \frac{1}{2}(I_c + I_n)$ // Mean intensity

$W_{avg} \leftarrow \frac{1}{2}(W_c + W_n)$ // Mean CB PI rate

$W_{avg}^s \leftarrow \frac{1}{2}(W_c^s + W_n^s)$ // Mean STE PI rate

2. Valence-Band Depletion Factor:

$D \leftarrow 1 - n_e/n_a$ // Fraction of neutrals still available

if $D < 0$ **then**

  $D \leftarrow 0$ // Clamp to lower bound

end

if $D > 1$ **then**

  $D \leftarrow 1$ // Clamp to upper bound

end

3. ODE Coefficients — Free-Electron Channel n_e:

$S_e \leftarrow W_{avg} D$ // Valence photoionization source

if $E_{ste} \neq 0$ **then**

  $S_e \leftarrow S_e + (W_{avg}^s + \beta_s I_{avg} n_e)\,(n_s/n_a)$ // STE→CB re-ionization: photo + avalanche

end

$L_e \leftarrow \beta_g I_{avg} D - \tau_r^{-1}$ // Avalanche gain minus trapping loss

4. ODE Coefficients — STE Channel n_s:

if $E_{ste} \neq 0$ **then**

  $S_s \leftarrow \tau_r^{-1} n_e$ // Gain from free-electron trapping

  $L_s \leftarrow -(W_{avg}^s + \beta_s I_{avg} n_e)/n_a$ // Loss by STE re-ionization (photo + avalanche)

else

  $S_s \leftarrow 0$ // No STE source if disabled

  $L_s \leftarrow 0$ // No STE loss if disabled

end

5. Exact Exponential Integration:

$n_e^{new} \leftarrow \text{EXACTEXPSTEP}(n_e, S_e, L_e, \Delta t)$ // Advance free electrons

if $E_{ste} \neq 0$ **then**

  $n_s^{new} \leftarrow \text{EXACTEXPSTEP}(n_s, S_s, L_s, \Delta t)$ // Advance STEs

else

  $n_s^{new} \leftarrow 0$ // Force zero STE density

end

6. Density Conservation Constraint:

if $n_e^{new} + n_s^{new} > n_a$ **then**

  $\text{scale} \leftarrow n_a/(n_e^{new} + n_s^{new})$ // Rescaling factor

  $n_e^{new} \leftarrow n_e^{new} \times \text{scale}$ // Rescale electron density

  $n_s^{new} \leftarrow n_s^{new} \times \text{scale}$ // Rescale STE density

end

7. Commit & Register Update:

$n_e[base + j + 1] \leftarrow n_e^{new}$ // Write electrons to global memory

$n_s[base + j + 1] \leftarrow n_s^{new}$ // Write STEs to global memory

$n_e \leftarrow n_e^{new}$ // Update local n_e register

$n_s \leftarrow n_s^{new}$ // Update local n_s register

$I_c \leftarrow I_n$ // Update local intensity register

$W_c \leftarrow W_n$, $W_c^s \leftarrow W_n^s$ // Update local rate registers

Algorithm 7: Two-population plasma rate-equation solver (Part 2/2): sequential trapezoidal time-stepping per radial point.

Function BUILDGRIDS *(Part 1/2: Grids & Propagation Operators)*:

Input: Configuration `cfg`, providing pump wavenumber k_0 (`komega`), carrier ω_0, and laser/material/grid parameters

Output: Intermediate grids and operators (assembled into `g` in Part 9)

1. Temporal & Spectral Grids:

$T_p \leftarrow$ `cfg.delta_t`$/\sqrt{2\ln 2}$ `// Intensity FWHM` $\rightarrow$ `field` $1/e$ `half-width`

$t_{max} \leftarrow 5\,T_p$ `// Window long enough for the pulse to vanish`

$\Delta t \leftarrow 2t_{max}/$`cfg.Nt` `// Uniform time step`

$t_{list} \leftarrow \text{Linspace}(-t_{max}, t_{max},$ `cfg.Nt`; endpoint=false$)$ `// Periodic time axis for FFT`

$f_{list} \leftarrow \text{FFTfreq}($`cfg.Nt`$, \Delta t)$ `// Frequency axis`

2. Radial Grid (Quasi-Discrete Hankel Transform):

$\{j_m\}_{m=1}^{N} \leftarrow$ Zeros of Bessel J_0 `// QDHT support points (`j_N`: last zero)`

$R \leftarrow$ `cfg.R_factor` $\cdot$ `cfg.w0` `// Spatial box radius`

$Y_{mn} \leftarrow \dfrac{2}{j_N}\dfrac{J_0(j_m j_n/j_N)}{J_1(j_m)^2}$ `// QDHT transform matrix (symmetric, self-inverse)`

$r_m \leftarrow j_m R/j_N$ `// Radial coordinates (`$r > 0$`)`

$\rho_m \leftarrow j_m/R$ `// Transverse spatial frequencies` $k_\perp$

3. Linear Dispersion Operator (Exact Sellmeier):

$\omega_{safe} \leftarrow \text{Clip}\big(\omega_0 + 2\pi f_{list},\ \frac{2\pi c}{5\,\mu m},\ \frac{2\pi c}{0.18\,\mu m}\big)$ `// Restrict to` $[0.18, 5]\,\mu$`m; avoid Sellmeier poles`

$n(\omega) \leftarrow \sqrt{1 + \sum_i B_i \lambda^2/(\lambda^2 - L_i^2)}$ `// Exact Sellmeier index,` $\lambda = 2\pi c/\omega_{safe}$

$k(\omega) \leftarrow \omega\, n(\omega)/c$ `// Wavenumber`

$k_1 \leftarrow \dfrac{k(\omega_0{+}\delta\omega) - k(\omega_0{-}\delta\omega)}{2\,\delta\omega},\quad \delta\omega = \omega_0/1000$ `// Inverse group velocity via central difference`

$\Delta k(\omega) \leftarrow k(\omega) - k_0 - k_1\,\Omega,\quad \Omega = 2\pi f_{list}$ `// Residual dispersion in co-moving frame`

4. Nonlinear Operators (Self-Steepening & Raman):

$\hat{T} \leftarrow 1 + f_{list}/f_0$ `// Self-steepening operator (`$f_0 =$ `cfg.frequency)`

$R(t) \leftarrow \Theta(t)\,\Omega_R^2\,\tau_s\,e^{-t/\tau_d}\sin(t/\tau_s)$ `// Causal Raman response (Heaviside` Θ`,` $t > 0$`)`

$R_f \leftarrow \text{FFT}\big(\text{IFFTShift}(R(t))\big)\,\Delta t$ `// Raman response in spectral domain`

5. 2D Space-Time Meshgrids:

$(tt, rr) \leftarrow \text{Meshgrid}(t_{list}, r_{list})$ `// (`r, t`) grids for pointwise operators`

$(\cdot, \rho\rho) \leftarrow \text{Meshgrid}(t_{list}, \rho_{list})$ `// Transverse-frequency grid`

Algorithm 8: Grid and operator construction (Part 1/2): temporal/radial grids, dispersion, nonlinear operators (`build_grids`).

Function BuildGrids *(Part 2/2: Plasma Constants, Absorber, Keldysh LUTs)*:

Input: `cfg` and Part 8 outputs (rr, R, k_0, ω_0)

Output: Dictionary `g` of precomputed grids, operators, plasma constants, and photoionization LUTs

1. Drude Plasma Constant & Gated Coefficients:

$\sigma \leftarrow \dfrac{k_0 e^2 \tau_c}{n_0^2 m_D\, \varepsilon_0\, \omega_0\,(1+\omega_0^2\tau_c^2)} \times 10^4$ `// Inverse Bremsstrahlung cross-section; ×10^4: m^2→cm^2 (m_D = cfg.meff_drude)`

if *cfg.enable_avalanche* **then**

  $\beta_g \leftarrow \sigma/U_i$ `// Valence avalanche coefficient`

else

  $\beta_g \leftarrow 0$ `// Avalanche disabled`

end

if *cfg.enable_ste* $\wedge$ *cfg.enable_avalanche* **then**

  $\beta_s \leftarrow \sigma/U_s$ `// STE re-ionization coefficient`

else

  $\beta_s \leftarrow 0$ `// STE avalanche disabled`

end

if *cfg.enable_recombination* **then**

  $\tau_r^{-1} \leftarrow 1/\tau_r$ `// Trapping/recombination rate`

else

  $\tau_r^{-1} \leftarrow 0$ `// Recombination disabled`

end

$\kappa \leftarrow \frac{1}{2} n_0 c\, \varepsilon_0 \times 10^{-4}$ `// Field-to-intensity factor (|E|^2 → W/cm^2)`

2. Boundary Absorber:

$\mathrm{mask}_r \leftarrow \exp\big(-(rr/0.9R)^{20}\big)$ `// Super-Gaussian radial absorber`

3. Keldysh Photoionization LUTs (Dual Channel):

$I_{cpu} \leftarrow$ Log-spaced intensity grid $[1,\ \sim 10^{17}]$ W/cm^2 `// Spline sampling nodes`

foreach *gap* $(U, LUT) \in \{(U_i, LUT_g),\ (U_s, LUT_s)\}$ **do**

  $W(I) \leftarrow \max\big(\mathrm{NaNtoNum}(\textsc{KeldyshRate}(\lambda, U, m_{eff}, n_0;\, I_{cpu})),\ 0\big)$ `// Evaluate & sanitize analytical rate`

  $f_{spline} \leftarrow \mathrm{PCHIP}(I_{cpu}, W)$ `// Monotone continuous interpolant`

  $\log I \leftarrow \mathrm{Linspace}(\log_{10} I_{min}, \log_{10} I_{max}, 4096)$ `// Uniform log-intensity LUT grid`

  $\mathrm{LUT} \leftarrow \mathrm{NaNtoNum}\big(|f_{spline}(10^{\log I})|\big)$ `// Sample onto GPU LUT array`

end

4. Return Grid Dictionary:

return `g` $\leftarrow \{t_{list}, r_{list}, Y, \Delta k, k_1, \hat{T}, R_f, rr, tt, \rho\rho, \sigma, \beta_g, \beta_s, \tau_r^{-1}, \kappa, \mathrm{mask}_r, \mathrm{LUT}_g, \mathrm{LUT}_s, \dots\}$ `// Precomputed state (+ LUT params log I_min, Δ_log^-1, I_min, I_max)`

Algorithm 9: Grid and operator construction (Part 2/2): plasma constants, boundary absorber, dual Keldysh LUTs (`build_grids`).

Function ENVELOPEGAUSSIANFOCUSED *(Initial Gaussian Envelope)*:

Input: Radial grid r, temporal grid t, configuration `cfg`, precalculated parameters dictionary `g`

Output: Complex spatiotemporal electric field envelope array $E(r,t)$

1. Beam Parameters Definition:

$T_p \leftarrow \texttt{cfg.delta\_t}/\sqrt{2\ln(2)}$ `// Conversion between Intensity FWHM and E-field 1/e half-width`

$C \leftarrow 1 + i\frac{2\cdot\texttt{cfg.begin}}{\texttt{g["b"]}}$ `// Complex beam curvature factor at initial propagation distance`

2. Field Construction:

$E(r,t) \leftarrow \frac{\texttt{g["E0"]}}{C}\exp\left(-\frac{r^2}{\texttt{cfg.w0}^2 C}\right)\exp\left(-\frac{t^2}{T_p^2}\right)$ `// Gaussian transverse profile and temporal envelope coupling`

return $E(r,t)$ `// Initialized spatiotemporal envelope`

Algorithm 10: Focused Gaussian Beam Initializer (`envelope_gaussian_focused`)

Class LinearOperator *(Linear Propagation Steps)*:

Method Constructor *(`__init__`): Operator Initialization*:

Input: Configuration object `cfg`, precalculated parameters dictionary `g`

Store base grids and arrays $k(\omega)$, Y, $j_{1,\text{last}}$, R, ρ, Δk as instance attributes

`// Model grids and operators caching`

$R_k \leftarrow R^2/j_{1,\text{last}}$ `// Forward QDHT normalization factor`

$R_k^{-1} \leftarrow j_{1,\text{last}}/R^2$ `// Inverse QDHT normalization factor`

Method HalfDiffraction *(`half_diffraction`): Transverse Spatial Propagation*:

Input: Complex field envelope $u(r, t)$, propagation step Δz

Output: Diffracted field envelope $u(r, t)$

1. Forward Hankel Transform:

$\tilde{u}(\rho, t) \leftarrow R_k \cdot (Y \cdot u)$ `// Transform to radial spatial frequency domain (QDHT)`

2. Diffraction Phase Shift:

$\tilde{u}(\rho, t) \leftarrow \tilde{u}(\rho, t) \cdot \exp\left(-i\frac{\rho^2}{2k(\omega)}\frac{\Delta z}{2}\right)$ `// Apply paraxial half-step diffraction operator`

3. Inverse Hankel Transform:

$u(r, t) \leftarrow R_k^{-1} \cdot (Y \cdot \tilde{u})$ `// Return to spatial domain`

return $u(r, t)$

Method HalfDispersion *(`half_dispersion`): Longitudinal Temporal Propagation*:

Input: Complex field envelope $u(r, t)$, propagation step Δz

Output: Dispersed field envelope $u(r, t)$

1. Forward Fourier Transform:

$\tilde{u}(r, \omega) \leftarrow \text{FFT}(u)$ `// Transform to spectral domain along temporal axis`

2. Dispersion Phase Shift:

$\tilde{u}(r, \omega) \leftarrow \tilde{u}(r, \omega) \cdot \exp\left(i\Delta k\frac{\Delta z}{2}\right)$ `// Apply chromatic dispersion half-step operator`

3. Inverse Fourier Transform:

$u(r, t) \leftarrow \text{iFFT}(\tilde{u})$ `// Return to time domain`

return $u(r, t)$

Algorithm 11: Linear Propagation Operators (`LinearOperator`)

Class NONLINEAROPERATOR *(Part 1/2: Setup & Matter Dynamics)*:

Method CONSTRUCTOR *(__init__): Operator Initialization*:

Input: Configuration object `cfg`, precalculated parameters dictionary `g`

Store scalar constants n_0, U_i, f_R, grids $\hat{T}, \tilde{R}_f$ and Keldysh LUTs as instance attributes `// Model grids and physical parameters caching`

1. Nonlinear Optical Coefficients:

if *cfg.enable_kerr* $\neq 0$ **then**

$\kappa_{\text{Kerr}} \leftarrow \frac{3 \cdot \texttt{cfg.chi3} \cdot \texttt{cfg.omega0}^2}{8 \cdot k(\omega) \cdot c^2}$ `// Derive Kerr self-focusing nonlinear coefficient`

else

$\kappa_{\text{Kerr}} \leftarrow 0.0$ `// Kerr effect computationally disabled`

end

2. Drude Plasma Coefficients:

$\kappa_{\text{pl}} \leftarrow \frac{\sigma(\omega)}{2} \cdot 100.0$ `// Inverse Bremsstrahlung absorption prefactor`

$C_{\text{pl}} \leftarrow 1 + i \cdot \texttt{cfg.omega0} \cdot \texttt{cfg.tau\_c}$ `// Complex Drude model plasma phase and absorption factor`

Method UPDATEPLASMA *(update_plasma): Electron Density Integration*:

Input: Field envelope u, densities ρ, ρ_s, time step Δt

Output: Updated plasma densities ρ, ρ_s (in-place modification)

1. Boundary Conditions:

$\rho(t = t_0) \leftarrow 0$ `// Initial free electron density at the front of the pulse`

$\rho_s(t = t_0) \leftarrow 0$ `// Initial Self-Trapped Exciton density at the front of the pulse`

2. Rate Equation Integration:

SOLVEPLASMARATEEQUATIONS$(u, \rho, \rho_s, \Delta t, \text{LUTs})$ `// Abstract CUDA kernel evaluating MPI, avalanche and STE dynamics`

return `// Densities are updated directly in device memory`

Algorithm 12: Nonlinear Operator: Initialization & Plasma Rate Equations

Class NonlinearOperator *(Part 2/2: Field Evolution RHS)*:

Method NonlinearRHS *(`split`): Evaluate Source Terms*:

Input: Complex field envelope $u(r,t)$, free electron density $\rho(r,t)$

Output: Tuple containing RHS derivative $\partial_z u$ and absorption map α

1. Photoionization and Absorption:

$I(r,t) \leftarrow |u|^2$ `// Calculate instantaneous spatiotemporal intensity profile`

$W_{\mathrm{PI}} \leftarrow \mathrm{EvaluateSpline}(I \cdot E_{\mathrm{ref}}^{-2})$ `// Calculate photoionization rate from Keldysh lookup table`

$D \leftarrow \mathrm{Clip}\left(1 - \frac{\rho}{\rho_{\max}}, 0, 1\right)$ `// Calculate bounded valence band neutral depletion factor`

$\alpha_{\mathrm{PI}} \leftarrow \left(W_{\mathrm{PI}} \cdot 10^6\right) \frac{U_i}{n_0 c \varepsilon_0 (I + 10^{-30})} \cdot D$ `// Derive nonlinear multiphoton absorption coefficient`

$\alpha_{\mathrm{total}} \leftarrow \mathrm{Max}(\kappa_{\mathrm{pl}}\rho + \alpha_{\mathrm{PI}}, 0)$ `// Combine plasma heating and MPI spatial absorption profile`

2. Kerr and Raman Delayed Response:

$I_{\mathrm{Kerr}} \leftarrow (1 - f_R) \cdot I + f_R \cdot \mathrm{Re}\left\{\mathrm{iFFT}(\mathrm{FFT}(I) \cdot \tilde{R}_f)\right\}$ `// Convolve intensity with delayed Raman molecular response`

3. Field Evolution (RHS):

$\tilde{N}_\omega \leftarrow \mathrm{FFT}\left(i\kappa_{\mathrm{Kerr}} I_{\mathrm{Kerr}} u - \alpha_{\mathrm{PI}} u\right)$ `// Compute nonlinear polarization source term in spectral domain`

$\mathrm{RHS} \leftarrow \mathrm{iFFT}(\tilde{N}_\omega \cdot \hat{T}) + \alpha_{\mathrm{PI}} u - \kappa_{\mathrm{pl}}(C_{\mathrm{pl}} - 1)\rho u$ `// Apply self-steepening operator and Drude plasma defocusing`

4. Vacuum Noise Suppression:

foreach *grid point where* $I < 10^{-30}$ **do**

  $\mathrm{RHS} \leftarrow 0 + 0i$ `// Force strict zero derivative in vanishing field regions`

end

return $\{\mathrm{RHS}, \alpha_{\mathrm{total}}\}$ `// Provide nonlinear derivative and combined diagnostic absorption`

Algorithm 13: Nonlinear Operator: Spatiotemporal Nonlinear Source Terms

Class INTEGRATOR *(Part 1/2: Setup & Main Loop)*:

Method CONSTRUCTOR *(__init__): Engine Initialization*:

Input: Configuration object `cfg`, Envelope definition `envelope`

1. Physical Grids & Operators:

$g \leftarrow$ BUILDGRIDS(`cfg`) `// Generate spatial, temporal and spectral discretized grids`

$\Delta z, \Delta t \leftarrow$ `cfg.dz`, g["dt"] `// Extract longitudinal and temporal integration steps`

$\hat{L} \leftarrow$ LINEAROPERATOR(`cfg`, g) `// Instantiate diffraction and dispersion solver`

$\hat{N} \leftarrow$ NONLINEAROPERATOR(`cfg`, g) `// Instantiate nonlinear polarization and plasma solver`

2. Fields & Energy Initialization:

$u(r,t) \leftarrow$ ENVELOPEINITIALIZER(r, t, `cfg`, g) `// Evaluate complex spatiotemporal initial envelope`

$\rho(r,t), \rho_s(r,t) \leftarrow 0, 0$ `// Initialize free electron and STE densities to zero`

$F_0(r) \leftarrow \sum_t |u(r,t)|^2 \cdot E_{\text{ref}}^{-2} \cdot \Delta t \cdot 10^4$ `// Integrate initial temporal intensity to get radial fluence`

$U_0 \leftarrow \sum_r F_0(r) \cdot 2\pi r \Delta r \cdot 10^6$ `// Integrate radial fluence for total initial pulse energy (`μ`J)`

3. Diagnostics History Arrays Allocation:

$N_{\text{save}} \leftarrow$ Floor(`cfg.nz`/`cfg.save_stride`) + 1 `// Number of longitudinal checkpoints`

Allocate memory for $F(z,r), \rho_{\max}(z,r), I_{\text{peak}}(z), E_{\text{dissipated}}(z)$ `// Initialize history arrays for macroscopic observables`

if *cfg.rho_t_stride* > 0 **then**

  Allocate subsampled arrays $\rho(z,r,t_{\text{sub}}), I(z,r,t_{\text{sub}})$ `// Initialize history arrays for spatio-temporal dynamics`

end

Method PROPAGATE *(propagate): Main Evolution Loop*:

Input: Configuration object `cfg`

Output: Dictionary of simulated physical diagnostics

foreach *step index* $i \in \{0, \ldots,$ *cfg.nz*$\}$ **do**

  1. Matter Dynamics Integration:

  $\hat{N}$.UPDATEPLASMA($u, \rho, \rho_s, \Delta t$) `// Integrate electron density temporally at current `$z$

  2. Optical Field Propagation:

  PROPAGATIONSTEP(Δz) `// Advance optical envelope spatially by `Δz

  $z \leftarrow$ `cfg.begin` $+ i \cdot \Delta z$ `// Update physical propagation distance`

  3. Diagnostics Checkpoint:

  if i mod *cfg.save_stride* $== 0$ **then**

    RECORDSTATE(z) `// Sample fluences, peak values, and densities`

  end

end

return FORMATRESULTS() `// Post-process fields and unfold symmetries for export`

Algorithm 14: Split-Step Fourier Integrator: Initialization & Main Loop

Class Integrator *(Part 2/2: Stepping Engine & Diagnostics)*:

Method PropagationStep *(`step`): Single Z-Step Advance*:

Input: Current field $u(r,t)$, plasma density $\rho(r,t)$, spatial step size Δz
Output: Updated field $u(r,t)$ evaluated at $z+\Delta z$ (in-place)

1. Linear Pre-Compensation:

$u \leftarrow \hat{L}.\textsc{HalfDiffraction}(u, \Delta z)$ `// Apply half-step paraxial diffraction (QDHT domain)`

$u \leftarrow \hat{L}.\textsc{HalfDispersion}(u, \Delta z)$ `// Apply half-step chromatic dispersion (FFT domain)`

2. Nonlinear Source Runge-Kutta 4 Integration:

$\_, \alpha \leftarrow \hat{N}.\textsc{NonlinearRHS}(u, \rho)$ `// Evaluate initial nonlinear multiphoton and plasma absorption`

$u \leftarrow u \cdot \exp\left(-\frac{\Delta z}{2}\alpha\right)$ `// Apply pre-RK4 symmetric absorption half-step`

$k_1, \_ \leftarrow \hat{N}.\textsc{NonlinearRHS}(u, \rho)$ `// RK4 Stage 1: Initial field derivative`

$k_2, \_ \leftarrow \hat{N}.\textsc{NonlinearRHS}(u + \frac{\Delta z}{2}k_1, \rho)$ `// RK4 Stage 2: Midpoint derivative prediction`

$k_3, \_ \leftarrow \hat{N}.\textsc{NonlinearRHS}(u + \frac{\Delta z}{2}k_2, \rho)$ `// RK4 Stage 3: Midpoint derivative correction`

$k_4, \_ \leftarrow \hat{N}.\textsc{NonlinearRHS}(u + \Delta z \cdot k_3, \rho)$ `// RK4 Stage 4: Endpoint derivative`

$u \leftarrow u + \frac{\Delta z}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ `// Advance complex field using weighted nonlinear source terms`

$\_, \alpha \leftarrow \hat{N}.\textsc{NonlinearRHS}(u, \rho)$ `// Evaluate final nonlinear absorption after RK4 update`

$u \leftarrow u \cdot \exp\left(-\frac{\Delta z}{2}\alpha\right)$ `// Apply post-RK4 symmetric absorption half-step`

3. Linear Post-Compensation:

$u \leftarrow \hat{L}.\textsc{HalfDispersion}(u, \Delta z)$ `// Apply half-step chromatic dispersion`

$u \leftarrow \hat{L}.\textsc{HalfDiffraction}(u, \Delta z)$ `// Apply half-step paraxial diffraction`

$u \leftarrow u \cdot \mathrm{mask}_r$ `// Apply absorbing boundary condition at radial edge`

Method RecordState *(`_record`): Observables Sampling*:

Input: Current longitudinal position z

$I(r,t) \leftarrow |u(r,t)|^2 \cdot E_{\mathrm{ref}}^{-2}$ `// Convert internal complex field to physical intensity`

1. Macroscopic Peak & Integrated Observables:

$F_{\mathrm{history}}(z_k, r) \leftarrow \sum_t I(r,t) \cdot \Delta t$ `// Calculate and store temporal fluence profile`

$\rho_{\mathrm{history}}(z_k, r) \leftarrow \max_t \rho(r,t)$ `// Extract and store peak free electron density`

$I_{\max}(z_k) \leftarrow \max_{r,t} I(r,t)$ `// Extract absolute maximum intensity on grid`

2. Spatio-temporal Sampling:

if $\mathit{cfg.rho\_t\_stride} > 0$ **then**

  $t_{\mathrm{sub}} \leftarrow t[$every `stride` steps$]$ `// Extract decimated temporal grid indices`

  $\rho_{\mathrm{history\_3D}}(z_k, r, t_{\mathrm{sub}}) \leftarrow \rho(r, t_{\mathrm{sub}})$ `// Store time-resolved plasma density profile`

  $I_{\mathrm{history\_3D}}(z_k, r, t_{\mathrm{sub}}) \leftarrow I(r, t_{\mathrm{sub}})$ `// Store time-resolved intensity profile`

end

$k \leftarrow k + 1$ `// Increment global checkpoint index`

Function RunSimulation *(run): Simulation Entry Point*:

Input: Laser parameters $(\lambda, w_0, \Delta t, E_{\mu J})$, material properties (n_2, U_i), numerical grid sizes (N_z, N_r, N_t), and integration boundaries $(z_{\text{begin}}, z_{\text{end}})$

Output: Dictionary of simulated macroscopic physical observables

1. State Configuration:

`cfg` $\leftarrow$ Config$(N_z, N_t, N_r, \lambda, E_{\mu J}, w_0, \Delta t, z_{\text{begin}}, z_{\text{end}}, n_2, U_i, \dots)$

`// Aggregate physical and numerical properties into global state`

2. Engine Instantiation & Execution:

solver $\leftarrow$ Integrator(`cfg`, `envelope`)

`// Initialize physical grids and split-step Fourier propagator`

results $\leftarrow$ solver.Propagate()

`// Execute spatial propagation and aggregate diagnostics`

return results

`// Return full history of converged fields`

Algorithm 16: Global Simulation Orchestrator (`run`)